

Operations Project Authorization Form

Approved at February 15, 2024 SDC

[Link to Meeting Minutes](#)

Project Title: Hyde Park Battery Energy Storage System	Project Number: 21334 (DSS)
Date Prepared: 01/12/2024	Company: Eversource EMA
Organization: System Planning	Class(es) of Plant: D SS
Project Initiator: Mariam Chatha	Project Category: Stations - Batteries
Managing Organization: System Planning	Schedule ID#: 80089926
Project Manager: Kelley, Adam	Project Type: Specific
Project Sponsor: Freeman, Lavelle A	Capital Investment part of original Op Plan: Yes
Estimated in service date: September 2027	Emergency Related Request? No
Eng./Constr. Resources Budgeted? Yes	Facility Type (check all that apply): ☐ PTF ☐ non-PTF ☐ local PTF ☒ Distribution ☐ N/A (General Plant)
EJ Community Impact: Yes	Authorization Type: Partial Funding
Total Capital Request: \$4.009M	

[1] Executive Summary

This is a partial funding request of \$4.009M to fund the preliminary engineering, siting, and permitting for the 10 MW/20 MWh utility-scale battery storage system (BESS) at 427 Cummins Highway near Hyde Park Substation #496 in Hyde Park, MA.

The primary objective of the proposed project is to help relieve the heavy loading condition of Hyde Park Substation #496.

Hyde Park Substation #496 is heavily loaded above 90% of the substation load carrying capability (LCC) and will continue to increase each year. A mid-term solution to bring additional capacity to the area is needed before a long-term solution (21192 - Hyde Park-Dorchester Area Supply Long-Term Plan, Partial Funding PAF was approved on August 14, 2023 but will be resubmitted to EPAC to accommodate for a 15 year station plan versus a 10 year one) will be in service. The expected in service date for the long-term project is December 2031. The proposed BESS is the proposed mid-term solution.

The project 22345 – Hyde Park-Dorchester Area Plan: Land Acquisition was approved by EPAC on September 11th, 2023 for the amount of \$6.275M and the property was purchased in October 2023. The property at 161-191 New England Avenue, Boston has been purchased to support project 21192 – Hyde Park-Dorchester Area Supply Long Term Plan.

This project had initial funding of \$200K approved in PowerPlan on June 16, 2021, and a revised 21334 - Hyde Park BESS – Battery Energy Storage System Solution, Rev 1 IFR for \$300K approved in PowerPlan on May 25, 2023. This Partial Funding PAF is being submitted concurrently with the project 21334 - Hyde Park BESS – Battery Energy Storage System Solution, Solution Selection Form).

The amount of \$4.009M will support the project up to the completion of preliminary engineering, at which point full funding will be requested.

[2] Project Costs Summary

Note: Dollar values are in thousands and rounded to nearest thousand
Project Number: 21334 Project Title: Battery Storage -
Hyde Park

D- Station		PAF (dollars in thousands)								Template version; rev25
Line item Category	Prior Authorized	Actuals to Date	2024 to Go	2025	2026	2027	2028	2029	Total	
1. ROW / Easements / Land purchases										
2. Environmental Approvals / Permits		\$6	\$156	\$161					\$323	
3. Outreach		\$14	\$13	\$13					\$39	
4. Siting Approvals / Permits		\$13	\$114	\$117					\$244	
5. Engineering / Design		\$30	\$169	\$174					\$374	
6. Materials (Eversource purchased)										
7. Construction (incl matl's by contractors)		\$14	\$11	\$11					\$36	
8. Testing / Commissioning										
9. PMT Project Mgmt. Team		\$70	\$154	\$158					\$382	
10. Removals										
11. Other		\$4							\$4	
12. Risks										
Project Directs Total w/Risks		\$151	\$616	\$635					\$1,401	
13. Indirects/Overhead		\$235	\$1,008	\$1,038					\$2,282	
14. AFUDC		\$33	\$86	\$208					\$326	
Project Total - Baseline Budget		\$418	\$1,710	\$1,880					\$4,009	
15. Contingency										
#N/A										
Project Total - Funding Approval Level	\$500	\$418	\$1,710	\$1,880					\$4,009	
16. Reimbursables/Customer Contribution										
#N/A										
Project Total (less reimbursables)	\$500	\$418	\$1,710	\$1,880					\$4,009	
Total O&M										
TOTAL REQUEST	\$500	\$418	\$1,710	\$1,880					\$4,009	

Detail other costs, project risk items and their allocations:

Risk	Allocation
1. *Risks will be accounted for in Full Funding Request	\$
2.	\$
3.	\$
4.	\$
5.	\$
6.	\$
7.	\$
Total	\$

Explain unique payment provisions, if applicable:

N/A

Identify and detail any unique O&M impacts resulting from this project:

Project will incorporate new equipment not normally maintained by Eversource including battery modules, inverters, and other energy storage system specific equipment. Contractual Language and requirements will be placed in the Battery Energy Storage System Vendor Request for Proposal to require training of Eversource personnel, material warranty and spares, and O&M manuals for all non-standard equipment.

[3] Technical Justification

Project Need, Objective and Justification

Project Need

Hyde Park Substation 2021 peak load was 157MVA, which exceeded the 2021 Load Forecast by 7MVA; It is 108% or 11MVA above Substation firm capacity of 146MVA; 98.7% of the Substation Load Carrying Capability (LCC) of 159 MVA. The firm capacity is the capacity based on the loss of a Hyde Park Substation transformer and the Load Carrying Capability is the firm capacity in addition to the amount of load that can be transferred to nearby Substations utilizing distribution circuit ties. In 2021, 7.7 MVA was transferred permanently off of Hyde Park Substation after the Substation peaked for that year. The 2022 post load transfer peak load was still 152MVA, 104% of the Substation firm capacity of 146MVA and 95.6% of the Substation LCC of 159 MVA. About an additional 5 MVA of load was transferred in 2022 after the Substation peaked for that year. In 2023, the substation was loaded to 97% of its firm capacity. A total of 13 MW of load was transferred to nearby Substations using existing field switches. However, even with these load transfers, the Hyde Park Substation is forecasted to be about 95% of the Hyde Park Substation LCC by 2024.

Load transfers are not a feasible option because they reduce the existing circuit transfer capability, increase the existing distribution lines' exposure, and reduce the spare capacity at the receiving Substation. Building a new Substation to address this thermal constraint is expected to take up to December 2031 at least and is a long-term solution to address the capacity constraint in the area. Therefore, a near term mitigation plan is needed to relieve this thermal Substation constraint until the long-term plan is in service.

Project Objective

The objective of the project is to alleviate Hyde Park heavily loaded condition, to prevent the Substation from being overloaded before the long-term solution, potentially a new Substation installation, can be materialized in the area.

Four (4) 2.5MW/5 MWh battery units will be installed at 427 Cummins Highway, each connected to a Hyde Park 13.8 kV distribution feeder. The batteries would be dispatched with a peak load shave functionality. They will charge during the light load and discharge during the peak load in order to lower the Substation peak. The Battery can operate through multi-day peak load events.

If there is any delay in relieving the Hyde Park area as defined in 21334 that has an in-service date of Q3 2027, the capacity to be delivered by the project will need to be delivered by other means. Delays may include but not limited to procurement, siting, final engineering, construction. Those options may include deployment of spot generation, voltage reduction, and load shedding during peak load hours typically from 6 PM to 8 PM. Based on the current 2023 load forecast, which may be subject to change in the future, if the BESS is not installed by the need date of 2027, two generators of one MW capacity would be needed in 2027 and 2028 and two generators of two MW capacity would be needed in 2029 until the BESS is installed. 8 MW of generation would be needed in the worst-case scenario where the BESS doesn't get installed

until the station is at 100% of LCC. Spot generation would be required to supply load during peak load hours and can remain on standby for the remainder of the time.

Project Scope

- The Hyde Park BESS project consists of a 10MW/20MWh energy storage system. The Project will be located at 427 Cummins Highway in Boston on Eversource owned property. The Hyde Park BESS will be operated in parallel with the Owner's electrical grid. The BESS shall be comprised of a system of battery racks, inverters, control system, pad mounted transformers, and switchgear. The BESS will be located and interconnected to Eversource's existing 13.8kV distribution infrastructure adjacent to the BESS facility.
- The BESS facility is proposed to interconnect to the following 13.8kV circuits supplied from Hyde Park substation:
 - 441-1401H
 - 362-1207H
 - 362-1209H
 - 496-H1XY
- Required Distribution System improvements include:
 - Extend distribution circuits 5,465' up to and into the Cummins Highway Property for the BESS interconnection. Scope of work includes three manholes, conduit, and associated cable.
- BESS Preliminary Scope of Distribution Work
 - Manhole Installation (3) concrete manholes with inside dimension of 6'x10'x8'
 - Conduit Installation (1,270') 6-5" PVC type EB encased in concrete from MH12362 to BESS via 3 new manholes
 - Primary Underground Cable Removal (13,065') of 15kV primary cable: 350/3, 3-350 FS, 3-500 FS, etc. (37 sections)
 - Primary Underground Cable Installation (16,395') of 3-700 FS 15kV primary cable (47 sections)
- Interconnection to Circuit 496-H1XY
 - Cut and remove X-Y main from MH18478 to MH12361 and replace with 2,950 feet of 3-700 FS (9 sections) and make joints.
 - Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.
 - Note: 1 Single three-way joint in MH18478 and 3 single multiple joints in MH12361.
- Interconnection to Circuit 441-1401H
 - Install new cable from MH12365 to MH12362 with 1,200 feet of 3-700 FS (3 sections) and make joints.
 - Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

- Interconnection to Circuit 362-1207H
 - Cut and remove cable from MH15729 to MH12361 (5 sections).
 - Install 1,810 feet of 3-500 FS (5 sections) from MH15729 to MH12361 and make joints.
 - Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.
- Interconnection to Circuit 362-1209H
 - Cut and remove cable from MH2434 to MH12358 (14 sections).
 - Install 5,355 feet of 3-700 FS (14 sections) from MH2434 to MH12358 and make joints.
 - Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

The \$4.009M partial funding being requested will be used to progress the project to the completion of Preliminary Engineering as well as the following:

Distribution System Planning

- Coordinate with the various stakeholders to carry the project up until full funding.
- Conduct PSCAD dynamic study using the specific battery inverter that will be procured for the project.

Transmission System Planning

- Perform transmission system impact study and obtain approvals of Proposed Plan Applications (PPA) from ISO-NE.

Capital Project Engineering

- Coordinate the overall engineering effort for the below external and internal disciplines.
 - Battery Energy Storage System Vendor
 - Substation Engineering
 - Protection & Control Engineering
 - Telecom Engineering
 - Distribution Engineering
- Finalize technical specifications for the Battery Energy Storage System Vendor Request for Proposal. Engineering has decided to bring on an engineering vendor to support this tech spec development.
- Procurement of Battery Energy Storage System vendor to develop conceptual and preliminary packages required for the installation and interconnection of the battery system as outlined above.

Project Management

- Overall project planning and scheduling
- Project reporting, document control and management, and financial projections
- Battery Energy Storage System Vendor procurement coordination

Siting

- Coordinate and lead siting effort which includes pre-filing meetings with the City of Boston for the Article 80 Small Project siting path. Coordination with Eversource legal on

the onboarding of outside legal counsel to assist with the submittal for Small Project Review, along with assistance throughout the duration of the permitting process and securing approval at the Boston Zoning Board of Appeals. Siting Filing will occur after Full Funding Approval.

Project Services

- Continue the community outreach and engagement strategy required for the project through the Project Services and Community Relations teams. This outreach strategy has been reviewed and communicated with the City of Boston and Boston Planning and Development Authority and has been aligned and coordinated with the outreach strategy for project 21192 – Hyde Park-Dorchester Area Supply Initiative.

Equity and Environmental Justice

- Identify EJ Communities and populations relevant to Project/Program. Identify languages spoken in stakeholder communities to enhance strategic communication with multilingual customers. Identify and communicate with communities and stakeholders in a broad geographic area to determine historical engagement and determine potential Project, Program, or policy impact.

Environmental

- Project support and preparation of regulatory agency required permits identified on the Sustainability and Environmental Affairs Department project review form.
- Coordinate required meetings with Massachusetts Department of Environmental Protection (MassDEP) for utilization of the Cummins Highway parcel for development as it is currently a capped landfill.

Background

In 2021, Hyde Park Substation exceeded the 2021 Load Forecast by 7MVA. It is 108% of Substation firm capacity of 146MVA, and 98.7% of Substation load carrying capability (LCC) of 159MVA. In 2022, 7.7MVA was transferred permanently off Hyde Park Substation. The 2022 post load transfer peak load was still 152MVA, 104% of the Substation firm capacity of 146MVA and 95.6% of the Substation LCC. In 2023, the substation was loaded to 97% of its firm capacity after additional load transfer.

Below is the Hyde Park 10 year load forecast and percentage of LCC.

	Station		2024	2025	2026	2027	2028	2029	2030	2031	2032
		Forecast Load	150.83	150.86	150.97	151.09	151.36	151.81	152.43	153.41	155
496	Hyde Park	Load/LCC	94.9%	95.5%	95.6%	96.2%	96.4%	97.3%	97.7%	98.3%	100%

A 13MW load transfer was implemented by using existing field switches. However, such transfer reduces the existing circuit transfer capability, increases the existing distribution lines' exposure, and reduces the spare capacity at the receiving Substation.

Additional capacity is required for Hyde Park to relieve the heavily loaded condition. Building a new Substation to address this thermal constraint will take a number of years and is a long-term solution to address the capacity constraint in the area. Therefore, a near term mitigation plan is needed to relieve this thermal Substation constraint until the long-term plan is in service Battery storage system or construction of distribution lines from nearby Substation to the area was considered.

[4] Alternatives Considered

Alternative 1 (preferred alternative):

Install four (4) 2.5MW/5MWH in total 10MW/20MWH Battery Storage system at Cummins Highway

With the proposed BESS installation and no distribution load transfers, the Hyde Park#496 load is expected to be relieved around 92% of the Substation LCC by 2028 as following:

	Station		2024	2025	2026	2027	2028	2029	2030	2031	2032
496	Hyde Park	Forecast Load	150.83	150.86	143.97	144.09	144.36	144.81	145.43	146.41	148
		Load/LCC	94.9%	95.5%	91.12%	91.78%	91.95%	92.83%	93.22%	93.85%	95.48%

The BESS is proposed to interconnect to four (4) 13.8kV distribution feeders, 496-H1XY, 441-1401, 362-1209H and 362-1207H. The proposed BESS system is expected to perform peak load shave functionality to relief the Substation heavy loading condition.

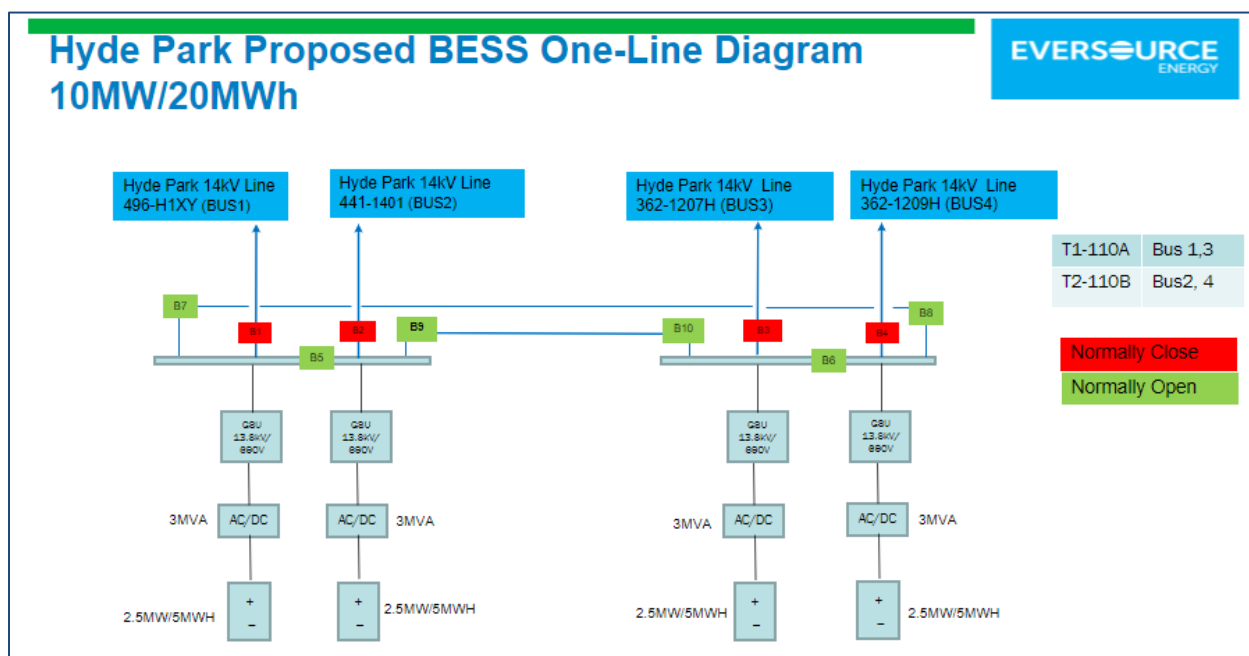


Figure1. BESS One line Diagram

Advantages:

- BESS can help relieve heavily loaded Hyde Park Substation.
- Construction is confined to a single site where the batteries can each interconnect to four different 13.8kV Hyde Park distribution feeders and provide substation bus diversity.
- Can be completed in a minimum of 3 years.
- This option has the least impact on the community.

- Shorter timeline for Outreach as compared to other alternatives since discussions with the city of Boston is already underway.
- Expected timeline for outreach, which has already commenced is approximately 12 – 18 months before a filing, which would likely be around Q2/Q3 of 2024 and continue ongoing thereafter.
- There has been early involvement with the city for this alternative, which helps to smooth the path for siting the BESS with city support and reduces opposition.
- After the long-term project is in service, the BESS can be redeployed to other locations where it can be useful.

Disadvantages:

- This alternative will likely require development review under Article 80 and a variance. If this is not permissible, a zoning exemption or relief may be required, which would take extra time in addition to the outreach time identified.
- If the Article 80E and Zoning Relief process is not successful, a 40A zoning exemption would be required from the DPU which would add significant time to the project.
- There has been local and neighborhood opposition for a proposed housing development on an adjacent parcel and similar opposition and concerns are expected for the BESS development. Project Services and Community Relations has been engaging with the City on this issue and will be performing targeted community outreach for this project and the Hyde Park solutions as a whole.

Hyde Park and surrounding downtown Boston parcels include areas where the majority of the population is a minority and speak English as a second language. Adding more capacity to the system helps provide reliable service to the population.

Hyde Park BESS Scope of Work is attached.

Conceptual (-25%/+50%) BESS Cost Estimate	\$44.379 million
Conceptual (-25%/+50%) Distribution Scope Cost Estimate:	\$ 4.793 million
Conceptual (-25%/+50%) Total BESS Cost Estimate	\$49.172 million total

Alternative 2:

Bring additional capacity to Hyde Park area by constructing two 13.8kV distribution lines from Needham Substation

The Needham station firm capacity is 142 MVA while the projected peak load in the next ten years is 88 MVA.

Based on an assumption that the new dedicated lines from Needham Substation#148 will run 100% underground path not currently occupied by common duct. The two new lines would run from Needham Station to nearby Hyde Park 13.8kV circuits, 496-H3 and 496-H6. The need to provide these two circuits would require:

19,600 feet of 9-6" PVC conduit

52 manholes

360 feet of 6-6" PVC conduit

350 feet of 2-5" PVC conduit
21 (55-foot poles)

The following cable work would be needed for the circuit that would relieve Hyde Park 13.8 kV circuit 496-H3:

19,900 feet of 3-700FS
2,450 feet of 477 ACSR
2 New Reclosers
2 New Underslung
2 new 45-foot poles

This new circuit would offload 7MVA from this 496-H3 circuit.

The following cable work would be needed for the circuit that would relieve Hyde Park 13.8 kV circuit 496-H6:

25,900 feet of 3-700FS
2,300 feet of 477 ACSR
2 New Reclosers
2 new Underslung

This new circuit would offload 6.21MVA from this 496-H6 circuit.

There is some load that would still be kept on 496-H6 and H3 just for cable monitoring purposes so that the cable will have readings and can therefore be utilized under a fault via a tie switch. If there is no load left on the feeders, they cannot be monitored to determine whether they can be utilized as distribution ties.

With this alternative, the total possible capacity on each circuit is 11.5MVA per circuit.

The below diagram shows the high-level route from Needham Substation to Hyde Park 13.8 kV circuits 496-H3 and 496-H6.

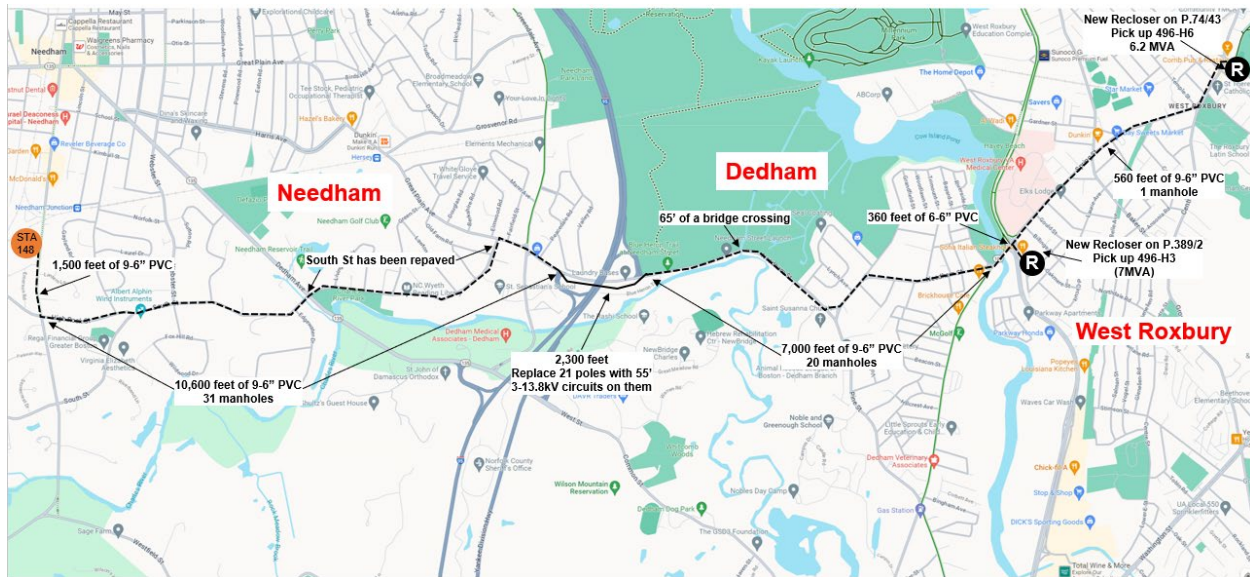


Figure 2 . Needham 13.8kV Feeders

Alternatively, the option of running the 13.8 kV circuits through an overhead right of way to connect to the two Hyde Park 13.8 kV circuits was also analyzed. This option was not feasible because the railroad easement along the path would limit the number of circuits that can run along the route and the width between the tracks and edge of easement is also too limited to accommodate two 13.8 kV circuits in some locations.

Advantages:

- This alternative would relieve the heavily loaded Hyde Park Substation.
- A filing with the Energy Facilities Siting Board (EFSB) or DPU is not anticipated, since this is a distribution related solution.

Disadvantages:

- Construction will be crossing multiple towns, busy highways and bridges, having an adverse impact on the community.
- This alternative will either take the same amount of time as alternative 1 or more.
- Due to the extensive in-street work required for this Alternative, Project Services and Community Relations would have to undertake substantial community outreach and dialogue with City officials, which has not commenced. No outreach has been done with municipalities on this alternative. Would require multiple meetings with city staff, elected officials, community-based organizations, etc. Expected timeline for outreach is approximately 12 – 24 months before a filing and continue ongoing thereafter. Extensive permitting would also be needed to cross multiple towns.
- Opposition in Needham is expected since a project was just completed through Needham and there is construction fatigue. If residents object or create organized opposition, significant delays to approvals are expected.
- Routing options would need to be discussed with cities and other stakeholders.
- Rights would need to be obtained from the towns/cities to install the conduits. Grant of location would be needed for Needham, Dedham and Boston.

- Permits would be needed to cross Route 128 and the bridges, and other entities such as MWRA or DCR. This can even take up to a year.
- The following caveats pose as disadvantages as well and can affect the cost estimates for this alternative:
 - South St in Needham has been recently paved, so it is uncertain whether curb to curb paving would be required.
 - Two overhead lines will be installed to cross Route 128. This may result in increased labor costs and decreases reliability of these circuits.
 - Conduit/pipes need to be installed over a bridge crossing at the Dedham/Needham line.
- Expect significant mitigation requests for roadwork restoration, such as curb-to-curb/gutter-to-gutter paving, ADA ramp restoration, new sidewalks, and/or traffic calming measures like speed humps, and other community benefits, which could include monetary payments for trees or other types of added green space.
- Depending on the extent of work required at the Needham and Hyde Park substations (not yet known), zoning relief may be required from the Town of Needham and the City of Boston.
- Based on the current forecast, this alternative would not remain as useful after the long-term project or new 115kV/13.8kV substation is in service to relieve load from Hyde Park and surrounding substations.

Alternative 2 Needham 13.8 kV Circuits are shown as an attachment.

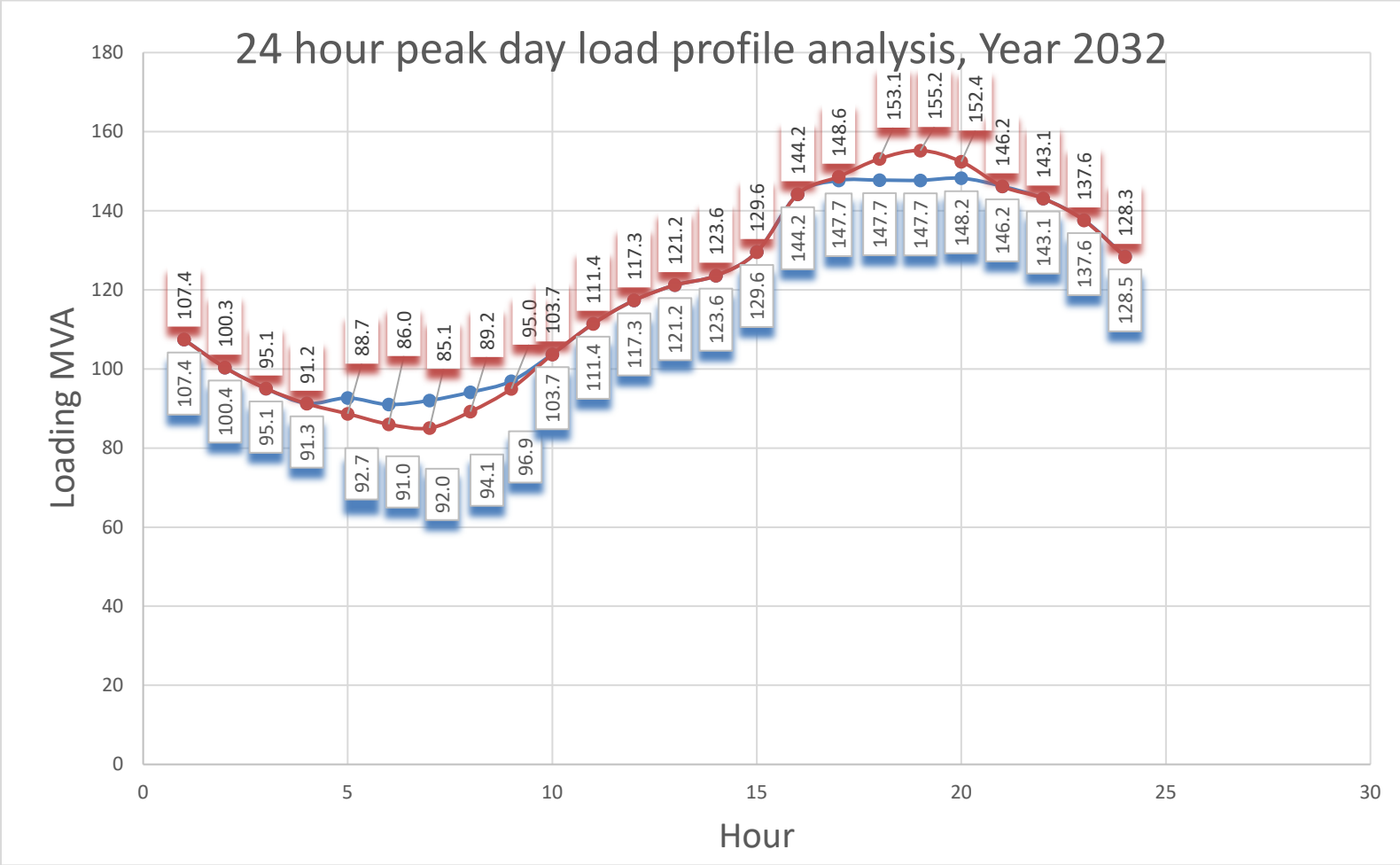
Conceptual (-25%/+50%) Cost Estimate:

\$46.473 million total

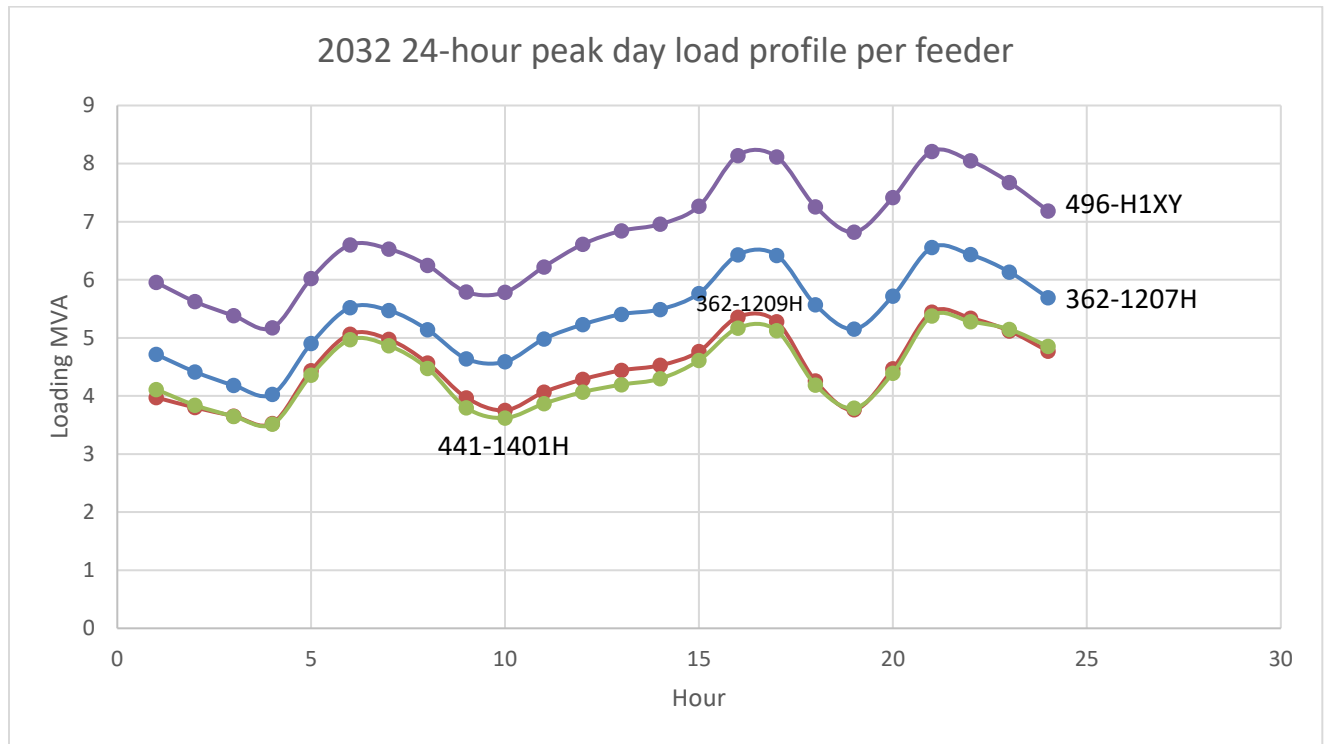
Hyde Park BESS Load Flow Analysis

The firm capacity of the Hyde Park Substation is 146 MVA while the LCC is 159 MVA. The LCC decreases incrementally by a slight amount every couple of years. The below graphs shows the 24 hour peak load day profile for the four selected feeders in MVA based on the 2032 modeled loads. The batteries act as a load and charge from 04:00 AM to 09:00 AM and discharge from 05:00 PM to 09:00 PM to decrease the Substation peak load.

Assuming the peak day load profile in the figure, the four (4) 2.5MW/5 MWh battery units can reduce the peak load of the Hyde Park Substation to approximately 95% of the LCC in year 2032 or approximately 148 MVA. The BESS would be interconnected to four (4) 13.8kV distribution feeders, 496-H1XY, 441-1401, 362-1209H and 362-1207H. One 2.5MW/5 MWh battery will be connected to each of the four feeders. The project is designed to meet the 10 year forecast, where the highest load being is in 2032.



In the above graph, the red line represents the Hyde Park Substation load without any batteries installed and the blue line represents the Substation load with the four batteries installed. Below graph shows the 2032 24-hour peak day loading on each of the four feeders that the BESS is interconnected to and assuming the BESS is operational.



As shown by the below table, the circuit loading is below 80 percent of the circuit ratings when the battery is charging from 5 AM to 9 AM and acting as a load.

	Feeder_ 362_ 1207H	Feeder_ 362_ 1209H	Feeder_ 441_ 1401H	Feeder_ 496_ H1XY
5:00 AM (MVA)	4.9	4.4	4.4	6.0
6:00 AM (MVA)	5.5	5.1	5.0	6.6
7:00 AM (MVA)	5.5	5.0	4.9	6.5
8:00 AM (MVA)	5.1	4.6	4.5	6.3
9:00 AM (MVA)	4.6	4.0	3.8	5.8
80% of Circuit Rating (MVA)	6.72	6.72	8.72	7.36
Circuit Rating (MVA)	8.4	8.4	10.9	9.2

During the annual planning forecast, System Planning will be monitoring and modifying the battery schedule based on the EV charging profile and its load on the Hyde Park distribution circuits.

The below graph shows the resultant load profile for the substation during the peak load day under contingency assuming 13.17 MVA of distribution load transfers and the battery is operational. This decrease of load to 141 MVA is due to a load transfer of 13.17 MVA which occurs during N-1 condition through distribution feeder ties and the decrease to 133.7 at 19:00 is due to the BESS.

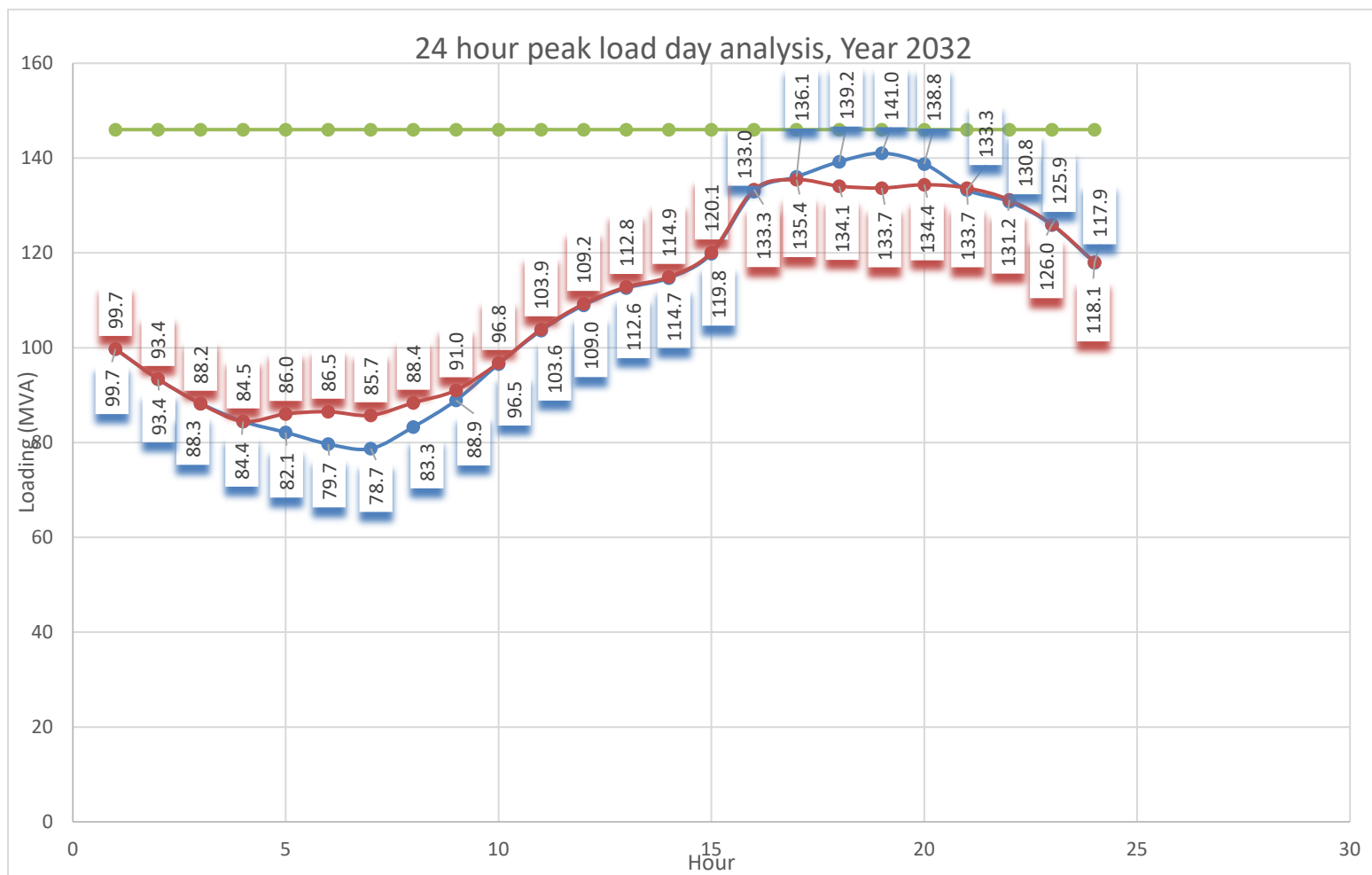
The red line represents the Hyde Park Substation load without any batteries installed and the blue line represents the Substation load with the four batteries installed.

Form Template Revised 12/07/2023

APS 1 – Project Authorization Policy Issued 12/31/21 Rev. 6

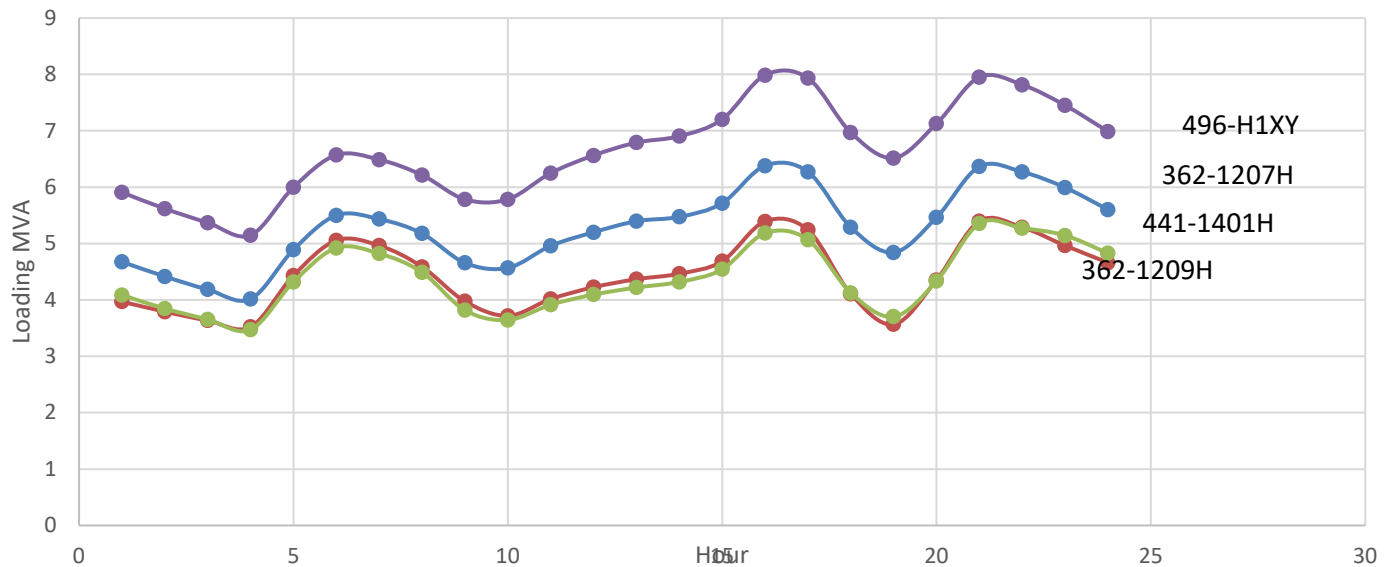
Policy Sponsor: EVP and CFO

JA-AM-2001-A Rev. 7



Below is a graph showing circuit loading for each of the four feeders under a Hyde Park Substation transformer contingency with the BESS connected to each of the four feeders. It shows the 24-hour peak load day profile for the four selected feeders in MVA based on the 2032 modeled loads. The batteries act as a load and charge from 04:00 AM to 09:00 AM and discharge from 05:00 PM to 09:00 PM to decrease the Substation peak load.

24hour peak day analysis per feeder



As shown by the below table, the circuit loading is below 80 percent of the circuit ratings when the battery is charging from 04:00 to 09:00 and acting as a load.

	Feeder_362_1207H	Feeder_362_1209H	Feeder_441_1401H	Feeder_496_H1XY
5:00 AM	4.89	4.43	4.32	5.996
6:00 AM	5.50	5.06	4.92	6.57
7:00 AM	5.43	4.96	4.82	6.49
8:00 AM	5.18	4.59	4.49	6.21
9:00 AM	4.66	3.97	3.82	5.783
80% of Circuit Rating (MVA)	6.72	6.72	8.72	7.36
Circuit Rating (MVA)	8.4	8.4	10.9	9.2

Hyde Park BESS Dynamic and Steady State Report is attached as attachment G.

[5] Project Schedule

Milestone	Scheduled Completion Date
IFR Approval	June 2021
SDC Approval (for applicable projects)	Feb 2024
Conversion of Tx development costs, 183 to 107 (applicable to projects that require SSF approval and/or ISO preferred alternative selection)	N/A
EPAC Partial Funding Approval (for applicable projects)	May 2024
Preliminary Design Completion (or latest pre-full funding design phase)	June 2025
EPAC Full Funding Approval	August 2025
Siting Approval (for applicable projects)	October 2026
Start of Construction	June 2026
Project In-Service Date (ISD)	September 2027
Project Need Date (for applicable projects)	Present
ISO Related Milestones: (for applicable projects)	
• ISO Preferred Alternative Selected	N/A
• PAC Presentation	N/A
• TCA Submittal	N/A

P6 schedule and Constructability Review are attached.

[6] Regulatory Approvals

Regulatory Approvals will be finalized after Full Funding.

- This alternative is expected to require Article 80 Small Project Review and a variance from the Boston Zoning Board of Appeal. Siting, with the assistance of outside legal counsel, would file for and obtain the required zoning relief.
- If the City is unsupportive of providing zoning relief, then Siting would file for a zoning exemption under M.G.L. Chapter 40A §3 from the Department of Public Utilities (DPU).
- The expected timeframe for a zoning exemption would be about 18 months to 24 months from date of filing to a decision.
- Necessary environmental permit approvals include:
 - Environmental Protection Agency (EPA) National Pollution Discharge Elimination System (NPDES) Construction General Permit (CGP) and Stormwater Pollution Prevention Plan (SWPPP)
 - Requires development/engineering of structural stormwater management system for facility and development of Operations and Maintenance (O&M) Plan for long term operation of stormwater management system, including semi-annual inspection, maintenance, and record keeping requirements
 - Requires regular construction phase erosion and sediment control inspections and reporting
 - Massachusetts Department of Environmental Protection (MassDEP) Landfill Post-Closure Use Permit (PCUP)
 - Requires both construction phase and long-term operation inspection, monitoring, and reporting requirements
 - May require installation of sophisticated gas monitoring system inside facility buildings to send real-time alert to Operations group for any buildup of combustible and/or toxic gasses within enclosed spaces
 - Soil and Groundwater Management Plan
 - Dictates regulatory requirements for construction phase handling, management, and disposal of contaminated soils and groundwater regulated under Massachusetts Contingency Plan
 - Spill Prevention, Control, and Countermeasure Plan (SPCC) and Tier II Reporting to EPA, MassDEP, MEMA, Local Fire Dept / Emergency Response Agency
 - Required for any new oil filled equipment with capacity >1,320 gal.
 - Required for onsite storage/use of large quantities of hazardous materials (batteries)

EJ Community Considerations:

	EJ Mapping	Identify EJ Communities and populations relevant to Project/Program, through creation of an EJ Map. EJ Map will identify, by Census Block: • Neighborhoods with English isolation • Low-income areas • Areas with concentrations of racial/ethnic minorities • Areas with vulnerable populations (including those with high existing environmental burdens) • Areas with combinations of any of the above • Distressed municipalities and any sub-areas with concentrations of EJ communities • Areas containing sensitive infrastructure or critical facilities (e.g., public housing, nursing/elder care homes or facilities, parks, hospitals, schools) • Areas containing high residency of low to moderate income renters
	Stakeholders Mapping & Engagement	Identify and communicate with communities and stakeholders in a broad geographic area to determine historical engagement and determine potential Project, Program, or policy impact, including local community-based organizations and leaders, state-wide NGOs and indigenous tribes.

	LANGUAGE TRANSLATION SERVICES	Identify languages spoken in stakeholder communities to enhance strategic communication with multilingual customers. Language identification data shall be used to inform and secure interpreters, generate appropriate translations of materials, conversations, dialogue, and in-community events.
	PUBLIC ENGAGEMENT	Host public engagement processes, including through non-traditional and community-based communication channels, to solicit public input. This may include, as applicable: • Determining the level of community interest • Conducting analysis of stakeholders who will be immediately impacted by the delivery or implementation of the program/project/services • Creating and implementing engagement & communication strategies • Hosting in person and or virtual public feedback sessions to clearly define program/project/services scopes of and time lines • Identify additional stakeholders to attend future public engagement activities and follow-up on feedback received

	Feedback	Establish a database to track all outreach events, interactions, materials, communications received from communities, individuals, and organizations; record all responses, all changes/modifications to projects or programs, and all remediation, mitigation, and community benefit arrangements made to or with respect to relevant EJ communities.
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[7] Risk and Risk Mitigation Plans

Contingency:

Contingency will be identified under the full funding development.

Risk Mitigation:

Risks will be identified under the full funding development.

[8] References

References
1. 21192 – Hyde Park -Dorchester Area Supply Initiatives Partial Funding PAF
2. 22345 – Hyde Park-Dorchester Area Plan
3. 21334 – Hyde Park BESS – Battery Energy Storage System Solution, Rev 1 IFR
4. 21334 – Hyde Park BESS – Battery Energy Storage System Solution, Solution Selection Form

[9] Attachments (One-Line Diagrams, Images, etc.)

Please make sure all attachments provided here are referenced above in the appropriate section of the PAF with the correct attachment number.

Attachment	Included in Submittal?
A. Cost Estimate	YES
B. P6 Schedule	YES
C. Constructability Review	YES
E. Hyde Park BESS Scope of Work Distribution	YES
F. Alternative 2 Needham 13.8 kV Circuits	YES
G. Hyde Park BESS Dynamic and Steady State Report	YES

EVERSOURCE

Project Authorization Form

INSTRUCTIONS:

It is the responsibility of the initiator to contact the subject matter experts in the listed area disciplines to determine if the project considerations contained in this list are applicable to their project. They should fill out the checklist and determine a transition plan for the purpose of project execution. See the PAF Guide for additional instruction.

Project Checklist - Transmission & Substation Capital Project	
Project Name: Hyde Park BESS – Battery Energy Storage Solution	Project Number: 21334
Facility Type: ☐ BPS ☐ BES ☐ PTF ☐ local PTF ☐ non-PTF ☐ CIP ☒ Distribution ☐ General Plant ☐ CONVEX ☐ SCADA	
PLANNING	
Is a NX-9 required?	No
Is an ISO-NE PAC presentation required?	No
ISO-NE Presentation Date (if completed):	N/A
Cost in ISO-NE Presentation (\$M) (if completed):	N/A
Is a PPA required?	Yes
ISO-NE Approval Date (if completed):	TBD
Is a TCA Application Required?	No
TCA Submittal Date (if submitted):	N/A
Cost in TCA Submittal (\$M) (if submitted):	N/A
Enter ISO-NE RSP / Asset Condition ID, if assigned:	ES-23-LSP-032
PLANNING/PROTECTION & CONTROLS	
Are RAS/SPS/UVLs affected?	No
OPERATIONS	
Outage Required?	☐ Primary Equipment (Power Transfer) ☒ Secondary Equipment (P&C only) ☐ Outage Not Required
Do SCLL Conditions Exist?	No
Has an outage schedule been approved?	No
Are Operations & Maintenance procedures/training required?	Yes
STANDARDS	
Does the project include standard equipment and designs?	Yes
SUBSTATION ENGINEERING	
Does this impact Revenue Metering	No
Is preliminary short circuit/ breaker duty analysis required?	No
Are there any changes to the baseline audible noise?	Yes
Is there an impact to the existing ground grid?	No
Is a Transient Over Voltage (TOV) analysis required?	Yes

EVERSOURCE

Project Authorization Form

Project Checklist - Transmission & Substation Capital Project	
Project Name: Hyde Park BESS – Battery Energy Storage Solution	Project Number: 21334
P&C ENGINEERING	
OP-22 - Are PMUs and DDR required?	No
If BPS, is an NPCC Directory #4 presentation required?	No
Are ISO-NE OP-24 Appendix B updates necessary?	No
TRANSMISSION LINE ENGINEERING	
Are there any changes that affect the baseline EMF?	No
Are there any changes that affect the baseline EMI?	No
SITING	
Has Siting reviewed this project?	Yes
Name of Siting contact:	Eamon McGilligan
Is a Siting filing required? <i>(If yes, list in regulatory approvals section)</i>	Yes
PERMITTING	
Is there any permitting required? <i>(If yes, list in regulatory approvals section)</i>	Yes
PROJECT SERVICES (OUTREACH) and ROW WORK COORDINATION	
Has Project Services reviewed this project?	Yes
Name of Project Services contact:	Meredith Boericke
What is the level of outreach expected?	High
Has ROW Coordination reviewed this project?	Yes
Is this project coordinating with other existing scheduled work?	No
ENVIRONMENTAL JUSTICE COMMUNITIES	
Has the Equity and Environmental Justice team reviewed this project?	Yes
Name of Equity and Environmental Justice contact:	Alicine Francois
Is any part of this project in an Environmental Justice Community?	Yes
Was the Equity Framework Checklist completed?	No
Has enhanced outreach required per State Regulatory requirements been completed? ¹	No
Enhanced Outreach will be developed by Full Funding.	
INITIATOR	
Has a field constructability review been completed?	Yes

Project Checklist - Transmission & Substation Capital Project	
Project Name: Hyde Park BESS – Battery Energy Storage Solution	Project Number: 21334
INVESTMENT RECOVERY	
Does the project require development of an Investment Recovery plan?	No _____
COST ESTIMATING	
What team/firm prepared the cost estimate?	Estimate was prepared by the Eversource Estimating Team _____
Name of the person who prepared the estimate:	Peter Grenier _____
Was the estimate reviewed by Eversource Estimating?	Yes - Full Review _____

Project Number:

Project Title:

D- Station

PAF (dollars in thousands)

And on years e version; rev25

Line item Category	Prior Authorized	Actuals to Date	2024 to Go	2025	2026	2027	2028	2029	Total
1. ROW / Easements / Land purchases									
2. Environmental Approvals / Permits		\$6	\$156	\$161					\$323
3. Outreach		\$14	\$13	\$13					\$39
4. Siting Approvals / Permits		\$13	\$114	\$117					\$244
5. Engineering / Design		\$30	\$169	\$174					\$374
6. Materials (Eversource purchased)									
7. Construction (incl matl's by contractors)		\$14	\$11	\$11					\$36
8. Testing / Commissioning									
9. PMT Project Mgmt Team		\$70	\$154	\$158					\$382
10. Removals									
11. Other		\$4							\$4
12. Risks									
Project Directs Total w/Risks		\$151	\$616	\$635					\$1,401
13. Indirects/Overhead		\$235	\$1,008	\$1,038					\$2,282
14. AFUDC		\$33	\$86	\$208					\$326
Project Total - Baseline Budget		\$418	\$1,710	\$1,880					\$4,009
15. Contingency									
#N/A									
Project Total - Funding Approval Level	\$500	\$418	\$1,710	\$1,880					\$4,009
16. Reimbursables/Customer Contribution									
#N/A									
Project Total (less reimbursables)	\$500	\$418	\$1,710	\$1,880					\$4,009
Total O&M									
TOTAL REQUEST	\$500	\$418	\$1,710	\$1,880					\$4,009

Breakout costs incl above

ST Labor		\$114	\$301	\$310					\$726
OT Labor									
Outside services - Labor		\$30	\$315	\$324					\$670
Materials (All materials incl. outside service purchased which is shown in constr. above)		\$2							\$2
Removals									
Other		\$4							\$4
Indirects		\$235	\$1,008	\$1,038					\$2,282
AFUDC		\$33	\$86	\$208					\$326
TOTAL CAPITAL REQUEST - with out Reimbursables	\$500	\$418	\$1,710	\$1,880					\$4,009

\$ 4,000,144

 Actual Work
 Critical Remaining Work
 Summary
 Remaining Work
 Milestone

Activity ID		Activity Name	Original Duration	Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTMEN	Resources	2024																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
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	A3500	TEST SERVICES - COC - Eversource Awards Contract - PO Issued	10	10	30-Oct-2025	13-Nov-2025	Not Started		PURC	Purchasing																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				

Hyde Park BESS - Battery Energy Storage System Solution 03-06-24										Full Activities										05-Jun-2024 11:39																					
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	A3280	Support Project/Testing & Execute test plan	217	217	30-Nov-2026	28-Sep-2027	Not Start		SUBTEC	LEAD CC																															
	BCN2200	System Commissioning	81	81	01-Jul-2027*	21-Oct-2027	Not Start		CONTR	O&M Sou																															
	80089926.09.03 LCE AUTHORIZATION (Remove)		21	21	19-Nov-2025	18-Dec-2025																																			
	A2750	Review and approve Outage Sequence Document (Sign-off)	1	1	19-Nov-2025	19-Nov-2025	Not Start		SUBTEC	LEAD CC																															
	A1730	LCE Review TOA / Applications (Pred=TOA#,Dwgs and Const Plan, has T18.01 successor	1	1	18-Dec-2025	18-Dec-2025	Not Start		SUBTEC	LEAD CC																															
	80089926.10 PROJECT CLOSE OUT		369	369	28-Oct-2026	27-Mar-2028																																			
	PCO1000	Project Close Out	369	369	28-Oct-2026	27-Mar-2028	Not Start																																		
	80089926.10.02 Electrical Close Out		124	124	06-Oct-2027	27-Mar-2028																																			
	EV1000	Prepare Electrical Field Mark Ups	21	21	06-Oct-2027	03-Nov-2027	Not Start	V-FLDAS	ENG.EN	LEAD CC																															
	EV1100	Eversource Engineering Prepares & Sends Electrical Field Mark Up Packge to Vendor	5	5	04-Nov-2027	10-Nov-2027	Not Start	ES-ASB	SS.ELEC	Shodhan																															
	EV1200	Vendor Engineering Verifies Electrical Mark Ups and Updates As-Built Drawing Package. Sub	30	30	11-Nov-2027	22-Dec-2027	Not Start	V-ASB	SS.ELEC	Shodhan																															
	EV1300	Eversource Engineering Verifies Quality of Submitted Electrical Drawings	20	20	23-Dec-2027	19-Jan-2028	Not Start	ES-ASB	SS.ELEC	Shodhan																															
	EV1400	Electrical Transmittal, Approve Final As-Built Package, and EDC Files and Issue Drawings to f	5	5	07-Feb-2028	11-Feb-2028	Not Start	ES-ASB	SS.ELEC	Shodhan																															
	PM1100	PM to Document Electrical As-Built Drawings Complete (Electrical, P&C, Communications)	1	1	14-Feb-2028	14-Feb-2028	Not Start	PM	PM	Adam S.																															
	COMPL1010	Final Inspection and Punchlist Work	90	90	22-Oct-2027	24-Feb-2028	Not Start		PM	Adam S.																															
	COMPL1020	Project Complete	0	0		27-Mar-2028	Not Start		PM																																
	80089926.10.03 P&C Close Out		124	124	06-Oct-2027	27-Mar-2028																																			
	PC1000	Prepare P&C Field Mark Ups	21	21	06-Oct-2027	03-Nov-2027	Not Start	V-FLDAS	ENG.EN	LEAD CC																															
	PC1510	Eversource Engineering Prepares & Sends Electrical Field Mark Up Packge to Vendor	5	5	04-Nov-2027	10-Nov-2027	Not Start	ES-ASB	P&C.P&C	P&C Eng																															
	PC1200	Vendor Engineering Verifies P&C Mark Ups and Updates As-Built Drawing Package. Submits	30	30	11-Nov-2027	22-Dec-2027	Not Start	V-ASB	P&C.P&C	P&C Eng																															
	PC1300	Eversource Engineering Verifies Quality of Submitted P&C Drawings	20	20	23-Dec-2027	19-Jan-2028	Not Start	ES-ASB	P&C.P&C	P&C Eng																															
	PC1310	P&C Transmittal, Approve Final As-Built Package, and EDC Files and Issue Drawings to Field	5	5	20-Jan-2028	26-Jan-2028	Not Start	ES-ASB	P&C.P&C	P&C Eng																															
	PC1500	FE Files P&C Design Drawings at Station	30	30	15-Feb-2028	27-Mar-2028	Not Start	ES-FLSA	P&C.P&C	P&C Eng																															
	80089926.10.04 Telecomm Close Out		112	112	22-Oct-2027	27-Mar-2028																																			
	COMM1000	Prepare Communication Field Mark Ups	21	21	22-Oct-2027	19-Nov-2027	Not Start	V-FLDAS	ENG.EN	Field Con																															
	COMM1100	Eversource Engineering Prepares & Sends Electrical Field Mark Up Packge to Vendor	5	5	22-Nov-2027	26-Nov-2027	Not Start	ES-ASB	SUBTEC	Bechir EI																															
	COMM1200	Vendor Engineering Verifies Communication Mark Ups and Updates As-Built Drawing Packag	30	30	29-Nov-2027	07-Jan-2028	Not Start	V-ASB	SUBTEC	Bechir EI																															
	COMM1300	Eversource Engineering Verifies Communication Mark Ups and Updates As-Built Drawing Pa	20	20	10-Jan-2028	04-Feb-2028	Not Start	ES-ASB	SUBTEC	Bechir EI																															
	COMM1400	Communications Transmittal, Approve Final As-Built Package, and EDC Files and Issue Draw	5	5	07-Feb-2028	11-Feb-2028	Not Start	ES-ASB	SUBTEC	Bechir EI																															
	COMM1500	Operations Files Communication Design Drawings at Station	30	30	15-Feb-2028	27-Mar-2028	Not Start	ES-FLSA	ENG.EN	Field Con																															
	80089926.10.11 LCE Closeout		5	5	29-Sep-2027	05-Oct-2027																																			
	LCE1000	LCE Review Test Data & Issue Commissioning Report (Preds=LCE Oversight of Test Contrct	5	5	29-Sep-2027	05-Oct-2027	Not Start		SUBTEC	LEAD CC																															
80089926.10.09 Work Order / Financial Close Out		10	10	28-Oct-2026	10-Nov-2026																																				
FINCO1000	PM/ Engr/ Plant Acct Develop Prelim UOP & UOR List(s)	5	5	28-Oct-2026	03-Nov-2026	Not Start	PM	PM	Adam S.																																
FINCO1010	Review UOP, compl forms, fwd to Plant Acctng, enter in wrkbk	5	5	04-Nov-2026	10-Nov-2026	Not Start		PM	Michael F																																
80089926.10.10 Project Management Close Out		40	40	09-Sep-2027	03-Nov-2027																																				
CSD1050	Construction Rep Finalize CASCADE Forms (within 14 days following ISD)	10	10	09-Sep-2027	22-Sep-2027	Not Start		PM	Adam S.																																
CSD1060	Asset Management Complete Asset Update(s) in CASCADE	30	30	23-Sep-2027	03-Nov-2027	Not Start		PM	Adam S.																																
80089926.13 NOT REQUIRED ACTIVITIES		680	431	18-Oct-2022	20-May-2026																																				
Actual Work Remaining Work Critical Remaining Work Milestone										Page 12 of 12										TASK filter: All Activities												© Oracle Corporation									



**Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review**

Project Title: Hyde Park Battery Energy Storage System	☐ Prior Constructability Review Performed?
	☐ Program Related Site-Specific Review?
Project Numbers(s): 21334	Project Constructability Phase: ☒ Conceptual Engineering Design ☐ Preliminary Engineering Design ☐ Detailed Engineering Design
Date(s) of Constructability Review: 10/31/2023	
07/20/2023	
Project Scope Statement: Installation of a 10MW/20MWh utility-scale battery storage system (BESS) along Cummings Highway near Hyde Park Substation #496 in Hyde Park, MA. This constructability review is being based off an initial project walkdown and tabletop review and is being completed prior to Conceptual Engineering for the BESS development. More encompassing reviews will be completed as the project progresses through engineering stages.	

Department Review/Walkdown Attendees*:

Required [X]	Attended [X]	Department	Name	Required [X]	Attended [X]	Department	Name
☐	☒	Project Management	A.Kelley	☐	☒	S/S Engineering	P. Fitzgerald
☐	☐	Asset Management		☐	☒	P&C Engineering	E. Revi
☐	☒	Constr. Management	R. Taves, J. Didomenico	☐	☒	Capital Projects Eng.	J. Zicko, L. Williams
☐	☐	S/S Technical / LCE		☐	☒	Telecom	B. Khoury
☐	☐	S/S Operations**		☐	☒	Environmental	C. Newhall
☐	☐	Field Engineering		☐	☐	Real Estate	
☐	☐	Line Engineering		☐	☐	ROW T&D Rights & Survey	
☐	☐	Vegetation Management		☐	☐	Survey Engineering	
☐	☒	Siting	Leah Zambernardi	☐	☐	Cost Estimating	
☐	☐	Safety		☐	☒	Project Services	E. Snedeker
☐	☐	SCADA		☐	☐	Material Procurement	

* Sign-off in the table above for the Detailed Engineering Design Constructability Review indicates that all Action Items/Concerns from prior Constructability Reviews have been mitigated/resolved as documented in this form. Not every department is required to attend walkdowns for every project but should be consulted for project impact considerations.

** Internal Eversource Substation Operations must be included in the review even if external construction resources are planned

[Project Manager] certifies that this review was performed to completeness, that all applicable departments were informed of the project and have had opportunity to provide input to this review:



Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review

6/5/24

Name / Signature

Date

Documents available for use prior to Constructability Review					
[X]	Document	[X]	Document	[X]	Document
☒	Initial Scope -Birdseye site plan -General Arrangement	☐	Preliminary Drawings -Civil Site Drawing -Electrical Drawing -Section Drawings	☐	Prints in hand for marking up -Panel Front/Read View -Control Enclosure Layout -Floor / Wall Plans
☐	Detailed Scope Document	☒	Proposed Schedule	☒	Construction Sequence/Timeline
☐	Previous Notes / Comments / Observations	☐	Major Material / Equipment List	☐	Replace / Reuse of equipment

Action Items and Resolutions:

Item No.	Const. Review Action Item/Concern	Design Resolution/Mitigation:
1		
2		
3		
4		
5		

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
A	Engineering & Asset Management		



Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
1	Scope of Work - Is scope adequately developed and sufficiently detailed for intended deliverable? - Any complimentary projects or other adjacent obsolete equipment for inclusion? - Any overlapping projects that could impact work/outages? - Consider Physical Installation, P&C Scope, Civil Scope, Engineering studies, Design standard adherence - Any remote station impacts? - Engineering Estimates for Design and Technical Support? (EE and EOC) Total Project	☐	Scope of work has been completed for both internal Eversource engineering and external Battery Energy Storage System vendor. Next steps following partial funding will be to progress engineering deliverables up to Preliminary Engineering where many of the items in the section will be addressed.
2	Materials Identified/Ordered -Any concerns with BOM/Major Materials? Review BOM lead times and project need dates -Specialty Major Materials: Identify Specialty Equipment & Materials potentials - temporary/replacement - long lead items? -Any Materials needed to facilitate construction access?	☐	Major material lead times for batteries, grounding transformers, and inverters are estimated to be at 12 months +/- . This duration has been built into the project schedule including standard procurement practices based on funding schedule.
3	Control House - Review Unmapped equipment interferences, Control/Battery Enclosure Layout, HVAC System (if equipment added), space for new or modified equipment, AC / DC station breaker availability and service loading, Panel Space, Expansion, one-line and/or riser type panelboard drawings, other.	☐	Indicative pricing effort has provided us with multiple equipment layouts for the site. Further engineering will be required to finalize layout, but the site is able to accommodate the installation of the equipment in multiple locations.
4	Ratings & Dimensions for Major Equipment. Any concerns upon field review? Equipment Rating One Line (and/or equipment thermal ratings tables) - Review for any errors in field	☒	No site concerns. As engineering progresses, we must ensure that the new access road can handle the equipment.



Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
5	Wall, Fence, Gate, and Foundation Considerations -Any permanent / temporary alterations or relocations required? -Pre-Project Sound Measurements Needed? Sound Walls? -Take measurements for Foundation quantities, dimensions and concrete volumes -CIP Review/Ballistic needs. Any Physical Security Requirement Consideration? -Assess condition of existing yard equipment foundations and viability for reuse (if proposed) -Review station expansion areas (if needed) for structures, old foundations, etc. -Any Steel Structural Plan Considerations? -Any impacts to construction plan?	☐	Site is currently fenced, and the future Battery Energy Storage System and associated equipment will also be fenced. Sound studies will be completed further along in the engineering process to ensure system adheres to dB requirements.
6	Below Grade / Enclosure Raceway - If possible, evaluate existing raceway and review spacing requirements. Review for any subsurface investigation needs or potential below grade obstructions. GPR/test pits needed? Geotechnical Investigation - is soil boring or resistivity investigation needed?	☐	See item 7 below.
7	Underground Obstructions -Drawing review concerns? -Is preconstruction or test excavation needed? -Is reuse of underground control cables / Conduits /replace Non-Shielded Cable feasible (if applicable) -Are spare conduits available? Are they useable? -Review if any areas will need to be plated for increased load bearing for equipment access (i.e., large Xfmr, etc.)	☐	Underground obstructions and conditions to be determined through required soil borings. It is understood that the site is a former landfill for construction debris and efforts will be made during the design process to limit excavation when possible.
8	Grounding Plan (Ground Grid) - Will footprint impact layout or other equipment? Identify locations to improve grounding. Impacts to foundations and ground grid plans. Observations on wetlands or ability to perform tests	☐	This is a new location, so a new ground grid will be installed in the location of the BESS. Coordination with the foundation plan and other equipment will be done to ensure the proper ground grid size. No site conditions pose any challenges for the ground grid.



Constructability Review Form: M6-PM-2012 - Eversource

Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
9	Nomenclature / Switching One-Line Diagram -Any obvious deviations from diagrams provided?	☒	No
10	BPS Separation Requirements - Is this BPS Substation? Y/N (part of PAF) - Changes to pre-existing BPS standards?	☒	No
11	IEC 61850 Substation Automation and Communication Requirements - Any Impacts/Retrofit needed?	☐	No
12	Any Upgrades to ES Standards - Is retrofit needed to bring up to ES standards? Can that be done?	☐	Capital Projects Engineering has written technical specifications used in the Battery Energy Storage System Request for Proposal. These technical specifications are being reviewed and cross referenced with a draft document that the Eversource Standards group has drafted.
B	Construction		
1	Any safety hazards specific to the facility? i.e., Low Bus, Deterioration, etc.	☐	No
2	Prior known Site-Specific Conditions -Common or Obvious station Issues -Missing scope items that could impact costs?	☐	Unknown subsurface obstructions are being carried as a risk until soil boring is completed.
3	Identify/Consider Required Depts for Construction? -Who is constructor: Internal vs External Resources -S/S Ops, GO, Test, Field Engineering, etc. -Specialty Contractors Required? -Required Oversight Needed - Estimated level of CR and LCE support needed	☐	Outside vendor will complete Construction of Battery Energy Storage System. Civil work for underground distribution circuit extension to be completed by outside vendor while cable pulling and termination to be completed by internal resources.



Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
4	Site Plan, Access, and Temporary Construction Facilities: -Confirm maintenance access for proposed new or replaced equipment -Verify clear footprint for yard equipment / Property Plan -General access to site, can areas outside substation be utilized? -Substation Entry Plan (driveway and drive path modifications) - Identification of Unique Requirements, i.e. (access bridges, station service, etc.) -Site Development Plan (grading and drainage) with Cut/Fill -Existing Rock Conditions? Potential Impacts (rock quantity)? -Any police details needed for deliveries, restricted hours, and/or need for extended hour approvals -Any Sanitary Civil Requirements? (Septic tanks, etc.) -Review turning radius for delivery trucks (if critical) -Any Substation Access and site/yard access Limitations? -Office Trailer Needs: Facilities, Utilities, Restroom, Container, Dumpster Space -Weather impacts to Site (Snow Placement) -Identify Material Laydown and Security -Where will new vs removed equipment go? -Construction permit for trailers outside substation fence?	☐	Layout requirements have accommodated for both maintenance clearances between equipment as well as around equipment and fence perimeter. Existing access road will be improved and utilized to access location. Further site and grading plans to be developed as engineering progresses as well as a pre-construction plan that will address the other items.
5	Unique Equipment Considerations: Gas insulated substation, Gas insulated line, Gas insulated bus, Pass Breaker, Statcom, Sync Condenser	☒	This is Eversource's second BESS installation, and this should still be considered a piece of unique/specialty equipment to be installed.



Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
6	Outage & Energization Plan and Construction Sequence: Review electrical working clearances for equipment during construction, Detailed look at protection zone changes, Locked sequence or random outages, Outage durations minimized. Considerations: -Approach Distances – Outage Requirements to maintain electrical clearances throughout proposed installation -Distribution Clearances -Clearance Holder Assignments -Drawings vs Field Conditions -Outage boundaries and interconnections -Phased approach needs for additional mobilizations -Verify outage sequence supports the planned construction sequence and timeline. (Lead Commissioning Engineer "LCE", Substation Operations, and Outage Planner input) -Any design alterations or considerations to Eliminate SCLL Conditions, Reduce Recall Time, Reduce Outage durations, Special Conditions -Additional outage requirements: Railroad Catenaries, Fiber outage, SCADA outage, Relay permissions -Seasonal or weather conditions impacts	☐	Will be completed further along in the engineering process.
7	Equipment Testing & Acceptance -Grounding plans / Resources? Will we need grounds hung? -Hi-pot testing and where will the equipment be set up?	☐	Will be completed further along in the engineering process.
8	Need for mobile generator and where connected (if needed)	☒	Mobile generators not required for construction. If construction is delayed then spot generation could be considered.
9	Need for mobile battery and where connected (if needed)	☒	Mobile batteries not required for construction. Mobile batteries could be considered as a partial solution to assist with schedule compression mitigation.
10	OT work required or Notable Permit Conditions (special hours)?	☐	TBD
11	Coordination of boundaries/seams (Identification of point of demarcation for	☐	Will be completed further along in the engineering process.



Constructability Review Form: **M6-PM-2012 - Eversource**
Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
	connection to existing equipment and/or for coordination with other municipalities)		
C	Environmental		
1	Soil Pre-Characterization or Grab Sampling Completed/Planned	☐	Soil borings and characterization planned.
2	Desktop investigation and EAD Review -Required Permits -Stormwater Considerations -Dewatering requirements -Wetland and buffer review	☐	EAD completed. Below are known permitting requirements at this time: -DEP Waste Management Program Post-Closure Use Permit for authorization to construct on closed landfill facility. -If soil is to be disposed offsite from Eversource property, soil samples will need to be taken to characterize the soil and a soil management plan will need to be developed to get approval from a disposal facility or landfill. - Requires annual Tier II reporting to MassDEP for batteries. - If project is expected to disturb ≥1 acre of land, then will require submittal of eNOI for coverage under the EPA 2022 Construction General Permit, incl. development of a SWPPP and associated inspections are required.
3	Risk or presence of PCB's/Asbestos/Lead/SF6	☐	Possible risk. Additional communication with DEP required along with review of landfill closure documents.
4	Transformer Spill Containment Recommended?	☐	Required spill containment will be determined with further engineering.
5	Expanding Footprint / New Buildings or future Expansion Plans? (Enviro Impact)	☐	Equipment will be ground mounted. A building will not be constructed.
6	Environmental oversight considerations? (Inspection type/Interval)	☐	Oversight will be required but length and frequency will be determined at permit application and approval stages.
7	Environmental mitigations - Any matting or seasonal needs? Review D&M Plan with CR.	☒	Not anticipated at this time.
D	Siting, Project Services, and ROW T&D Rights & Survey		
1	Any regulatory concerns or approvals needed? - Any concerns noted that could be mitigated by scope/design alterations? - Likely to require visual screening? What considerations (feasibility or location)? -Zoning Relief and Permits required?	☐	Views of BESS from abutting neighborhood and concerns for losing open land could be mitigated by scope/design alterations. Visual screening will likely be required, siting BESS closer to shopping plaza & away from neighborhoods. Screening/buffering likely required for abutting residential neighborhood. Zoning Relief and Permits are required. City of Boston Article 80 & Zoning Variance.



Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
2	Other siting considerations?	☐	Neighborhood is concerned about losing open land. Also interested in green energy and energy savings solutions.
3	Neighbor/Abutter impacts: - Any noise, visual impacts, or sensitive abutters - Any changes to lighting or additional lighting needed?	☐	Visual impact of equipment to be reviewed with City of Boston and community and accommodations to be made during engineering phase for any required screening of faux façade.
4	Review for roadway, gate access, trailers, equip height, noise, weekend/night work, or any potential public disturbance related to work	☐	Increased traffic will be seen during construction. Post construction, minimal traffic can be expected to the location for routine maintenance purposes.
5	Installation within Eversource Rights - Is Rights and Restrictions Review required?	☐	Eversource is owner of property
6	Expanding Footprint / New Buildings / Fence line or future Expansion Plans?	☐	Location of Battery Energy Storage System to take into consideration future development of the parcel for other electrical needs.
7	Proposed Equipment on same property? - Any Encroachments	☐	No encroachments anticipated
F	Protection & Controls		
1	Panel Graphic additions and/or Modifications	☐	Not applicable
2	Space on existing panels for new equipment? (front & rear of panel fronts)	☐	Not applicable
3	Panel fronts conform to drawings	☐	Not applicable
4	Access to bring new panels into control enclosure	☐	Not applicable
5	Nameplate list and location of nameplates	☐	Will be provided as engineering progresses
6	Verifies that relays and meters that will be an interconnected project work are as shown on the drawings.	☐	As engineering progresses drawings will be reviewed to ensure proper relaying and autotransfer scheme is achieved to allow the BESS to be supplying energy
7	Temporary protection, relays, control cable needed? How long will they be required?	☐	None anticipated at this time.
8	Will relay setting changes be needed? Any additional outage requirements for changes?	☐	At this stage of the engineering, it is not a requirement to do any additional relay setting changes as this will be interconnecting onto an existing distribution circuit. As the engineering progresses, the impact of the BESS interconnection on existing protection schemes will need to be assessed to determine the need for any remote setting changes.



Constructability Review Form: **M6-PM-2012 - Eversource**
Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
G	Telecom Engineering		
1	Review telecom equipment in existing control enclosure	☐	Not Applicable
2	Review path to get telecom circuits into the station and paths within the control house. Determine whether new underground duct or overhead tray work is needed.	☒	Fiber exists at the station and ADSS is available at 427 Cummins Highway if required. Wired or wireless options will be reviewed during the engineering phases.
3	Review current or proposed Telecom equipment power connections and validate with P&C any need for system 1 and system 2 separation.	☐	No need for system separation
4	Review current Telecom fiber diversity and validate with P&C any fiber diversity requirements.	☐	
5	Review and list any remote station impacts/updates	☐	Not Applicable
6	Confirm whether correct interface equipment is available (i.e.) for fiber termination or multiplexor termination	☐	This will be a new installation, and equipment will be specified to be installed
7	Confirm whether existing communications cabinets are enough to support additional telecommunications equipment installed	☐	New installation, will require new cabinets
8	Review space for any required additional communications cabinets which may be required to support additional telecommunications equipment installed	☐	New installation, will require new cabinets
9	Review third party Telecommunications at substation and confirm whether additional capacity or Ground Potential Rise protection is required	☐	

Photos

(Ex. Equipment control cabinet, Breakers, MODs, CCVTs, PTs, Transformers)

BESS High Level Scope

- Mid-term alternative before long-term solution is implemented
- Located off Cummings Hwy in Hyde Park

	Quantity	Total
Battery Size	1 installation	10 MW / 20 MWh
Inverter Size	6 x 2.5 MVA inverters 2 x 7.5 MVA sets	15 MVA



☐ **Manhole Installation**

- (3) concrete manholes with inside dimension of 6'x10'x8'

☐ **Conduit Installation**

- (1,270') 6-5" PVC type EB encased in concrete from MH12362 to BESS via 3 new manholes

☐ **Primary Underground Cable Removal**

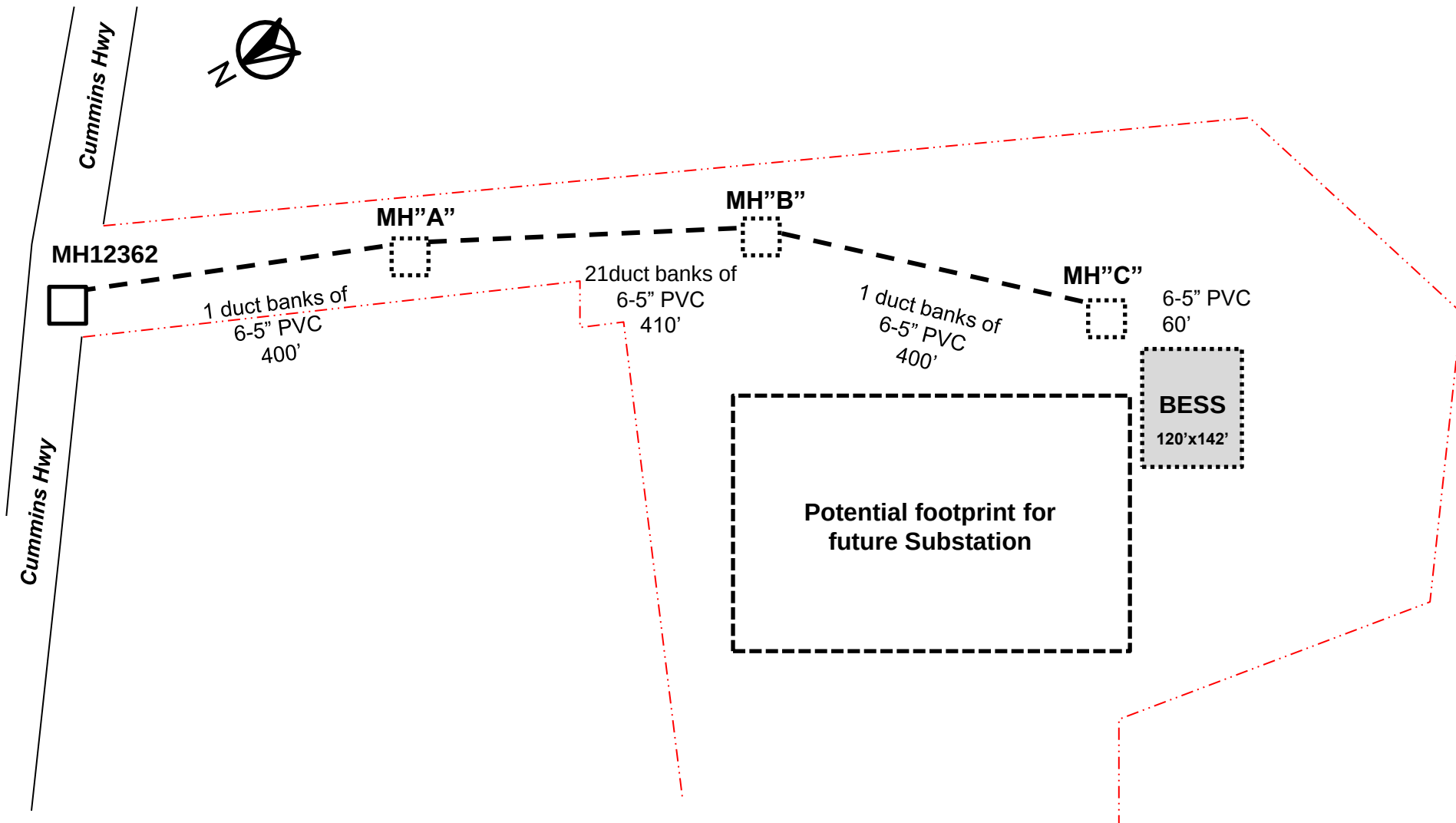
- (13,065') of 15kV primary cable: 350/3, 3-350 FS, 3-500 FS, etc. (37 sections)

☐ **Primary Underground Cable Installation**

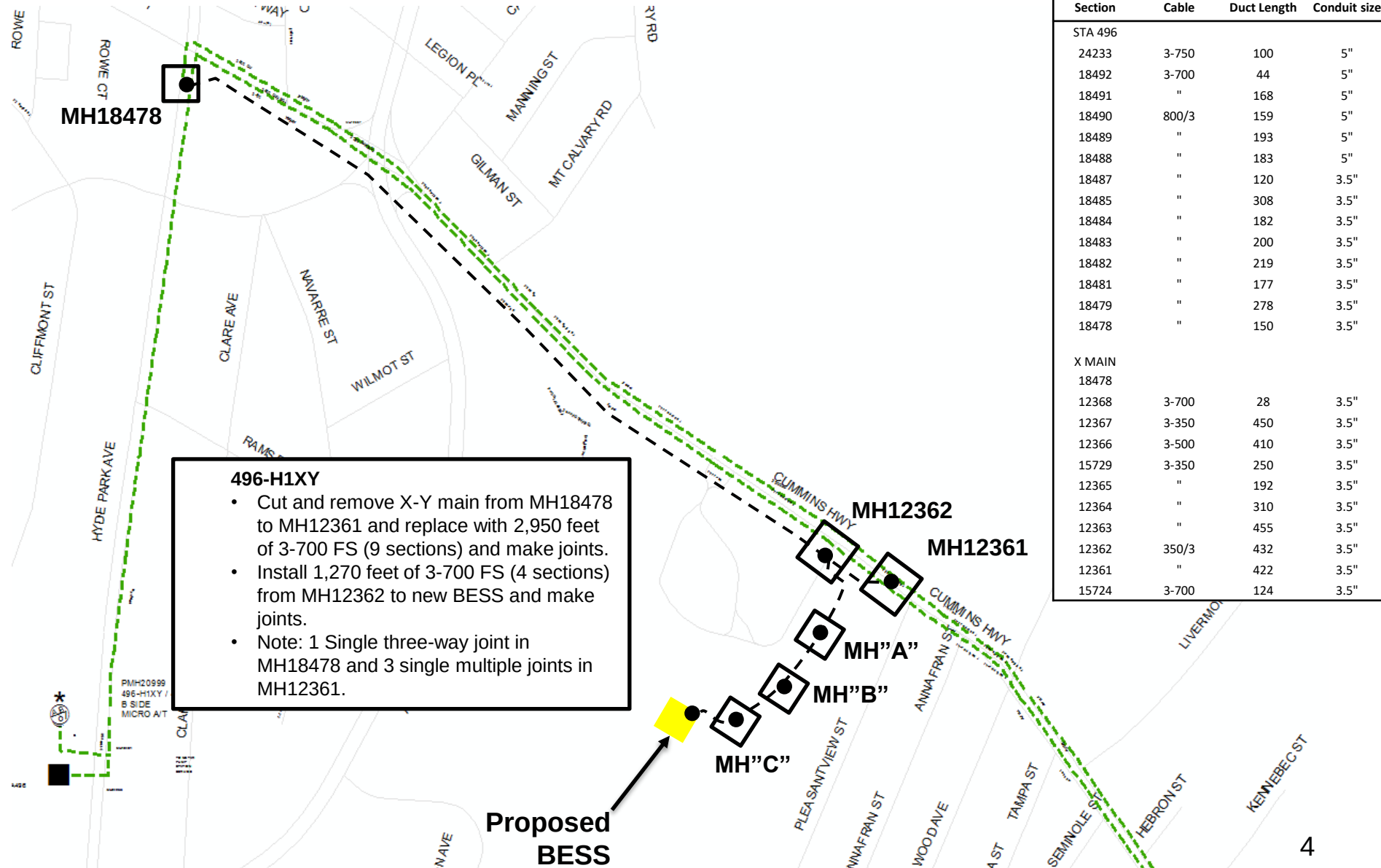
- (16,395') of 3-700 FS 15kV primary cable (47 sections)

Total Cost Estimate = \$4,459,417.00

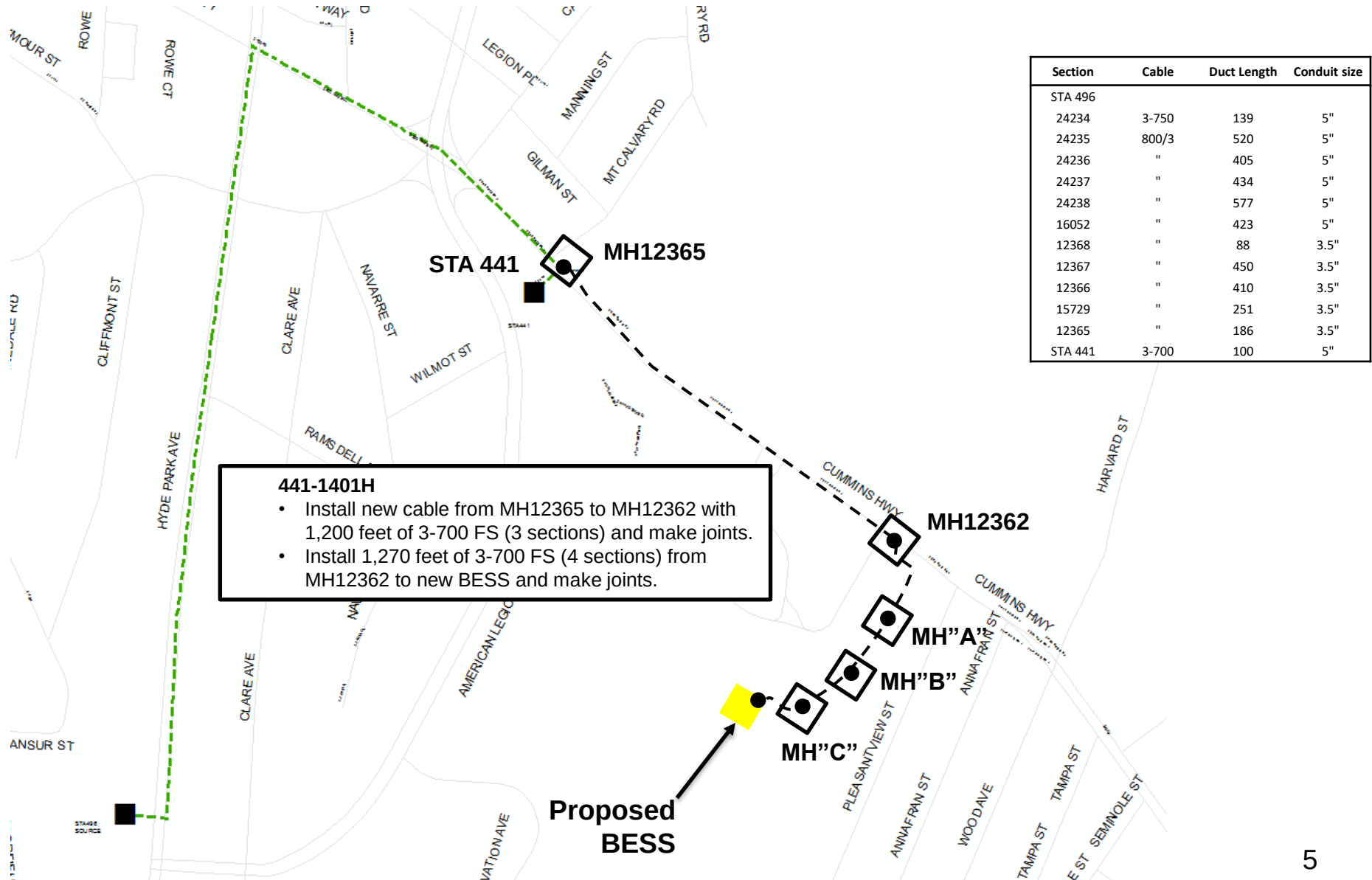
BESS Preliminary Design



BESS – 496-H1XY (Bus 1)



BESS – 441-1401H (Bus 2)

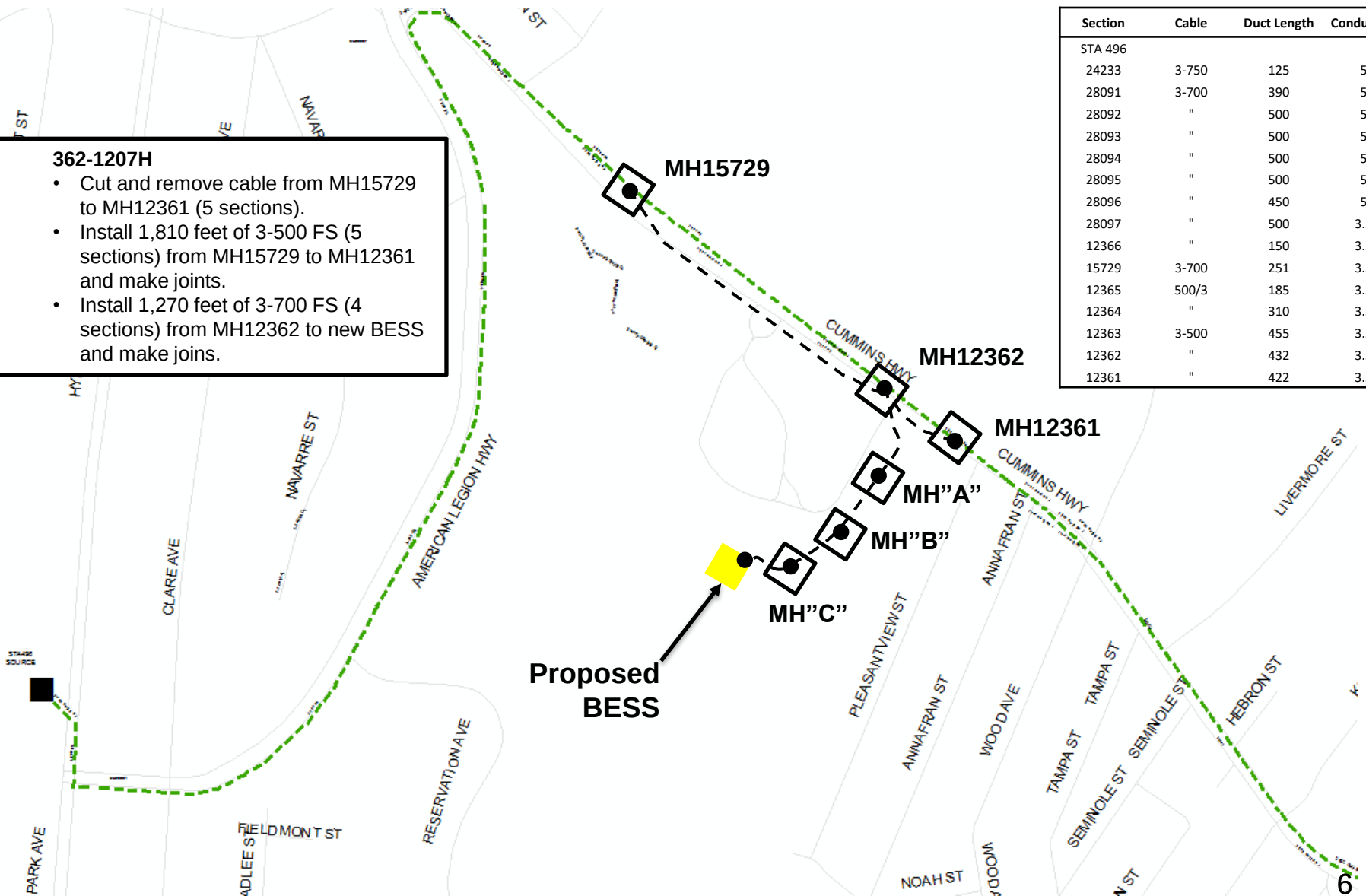


BESS – 362-1207H (Bus 3)

362-1207H

- Cut and remove cable from MH15729 to MH12361 (5 sections).
- Install 1,810 feet of 3-500 FS (5 sections) from MH15729 to MH12361 and make joints.
- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

Section	Cable	Duct Length	Conduit size
STA 496			
24233	3-750	125	5"
28091	3-700	390	5"
28092	"	500	5"
28093	"	500	5"
28094	"	500	5"
28095	"	500	5"
28096	"	450	5"
28097	"	500	3.5"
12366	"	150	3.5"
15729	3-700	251	3.5"
12365	500/3	185	3.5"
12364	"	310	3.5"
12363	3-500	455	3.5"
12362	"	432	3.5"
12361	"	422	3.5"

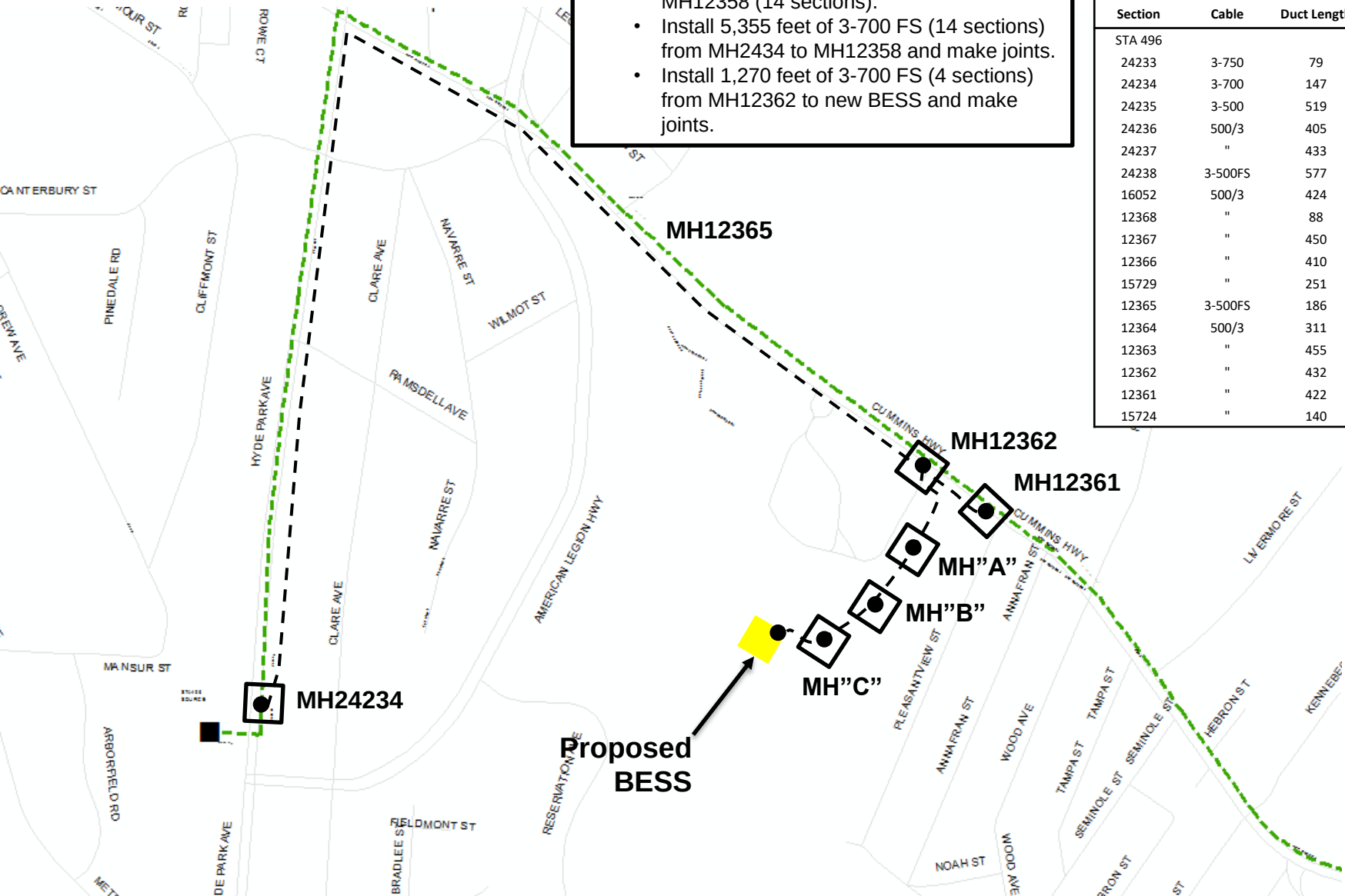


BESS – 362-1209H (Bus 4)

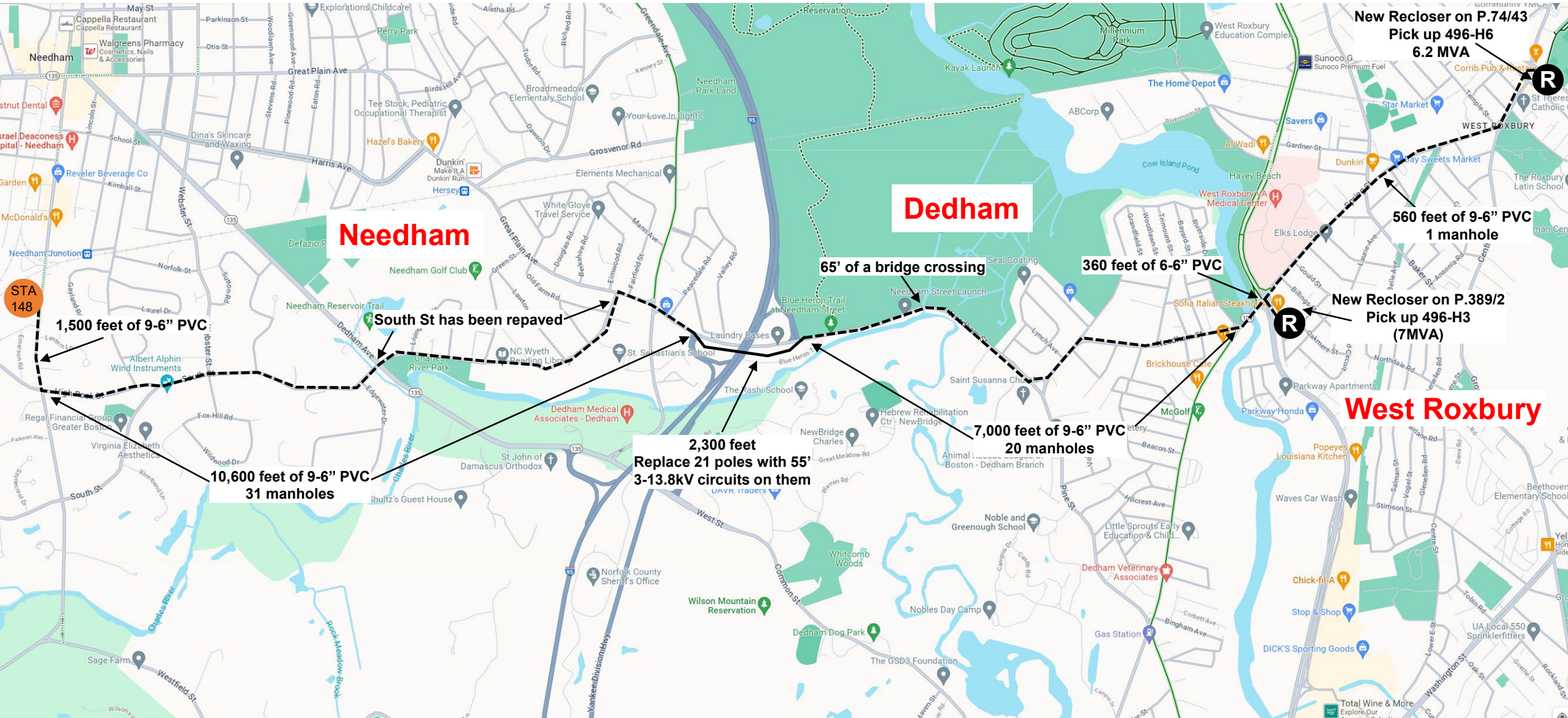
362-1209H

- Cut and remove cable from MH2434 to MH12358 (14 sections).
- Install 5,355 feet of 3-700 FS (14 sections) from MH2434 to MH12358 and make joints.
- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

Section	Cable	Duct Length	Conduit size
STA 496			
24233	3-750	79	5"
24234	3-700	147	5"
24235	3-500	519	5"
24236	500/3	405	5"
24237	"	433	5"
24238	3-500FS	577	5"
16052	500/3	424	5"
12368	"	88	3.5"
12367	"	450	3.5"
12366	"	410	3.5"
15729	"	251	3.5"
12365	3-500FS	186	3.5"
12364	500/3	311	3.5"
12363	"	455	3.5"
12362	"	432	3.5"
12361	"	422	3.5"
15724	"	140	3.5"



Conduit and Route from STA 148 to 496-H3 & 496-H6



Scope of Work to relieve STA 496 via STA 148.

19,600 feet of 9-6" PVC conduit

52 manholes

360 feet of 6-6" PVC conduit

350 feet of 2-5" PVC conduit

21 (55 foot poles)

To Relieve 496-H3 (7 MVA)

19,900 feet of 3-700FS

2,450 feet of 477 ACSR

2 New Reclosers

2 New Underslung

2 new 45 foot poles

To Relieve 496-H6 (6.2 MVA)

25,900 feet of 3-700FS

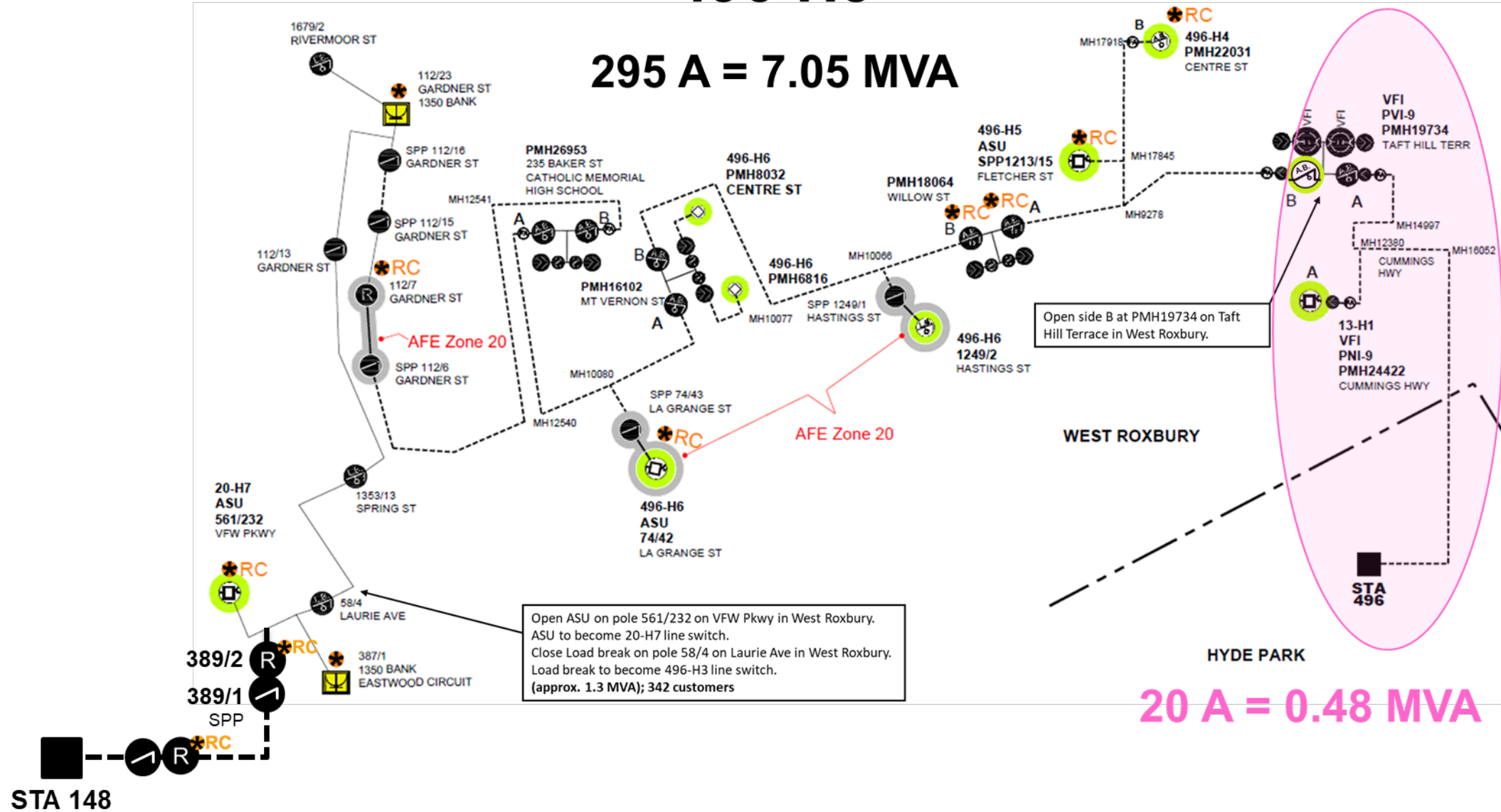
2,300 feet of 477 ACSR

2 New Reclosers

2 new Underslung

496-H3

295 A = 7.05 MVA

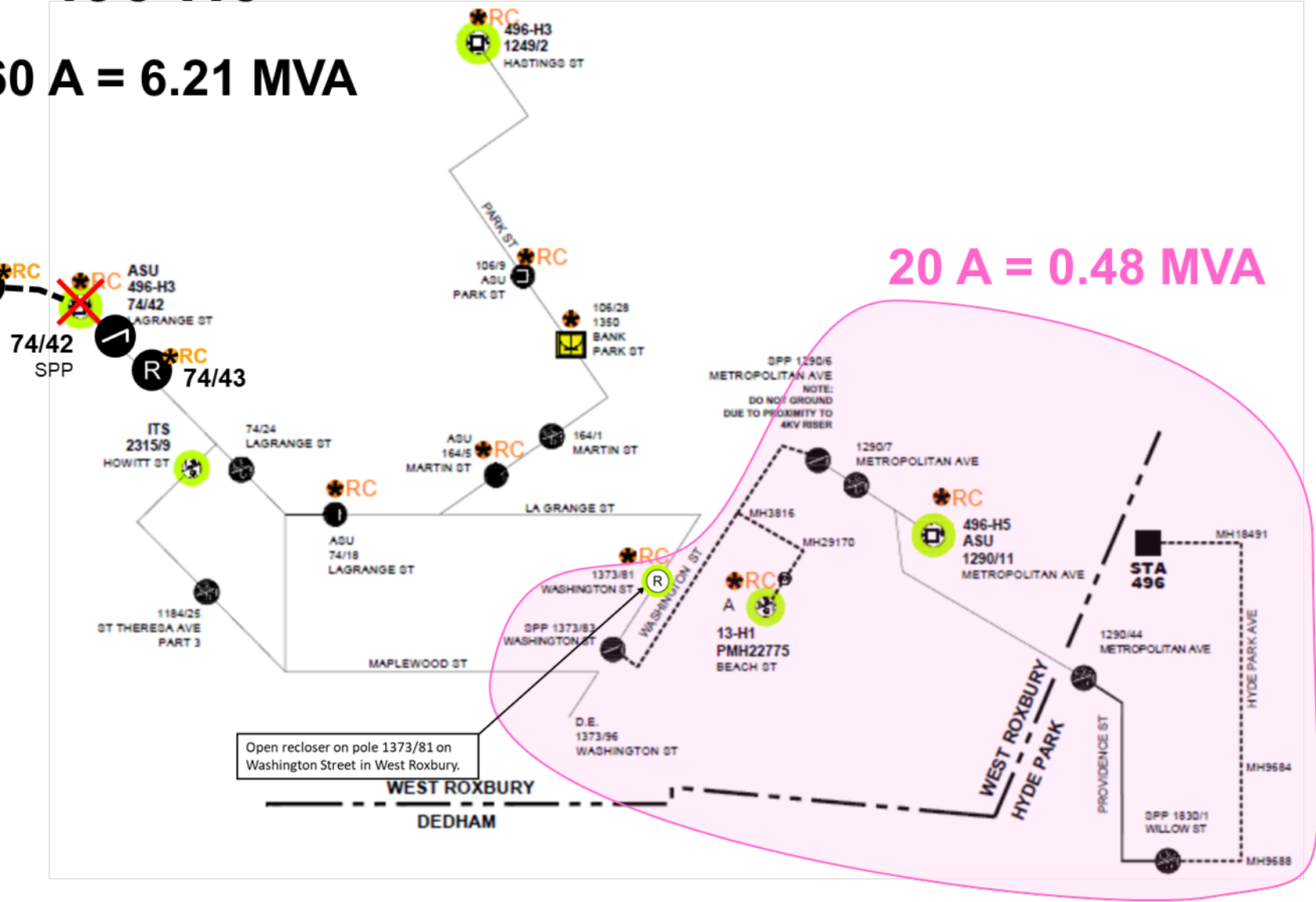


496-H6

260 A = 6.21 MVA

STA 148

20 A = 0.48 MVA





Hyde Park Substation Battery Energy Storage System (BESS)

Distribution System Impact Study: Steady State and
Dynamic Analysis

Report

System Planning MA

Prepared By: Mariam Chatha and Souresh Mukherjee

Revision History

Date	Rev.	Description
	1	Report

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Executive Summary

This report includes the dynamic analysis performed for the Battery Energy Storage System (BESS) being interconnected at four 13.8 kV Hyde Park distribution feeders at Cummins Highway in Boston on Eversource owned property. The four feeders are 496-H1XY, 441-1401, 362-1209H and 362-1207H supplied from Hyde Park Substation. There will be a 2.5MW/5MWH battery on each of the four feeders making the project consist of a total of 10 MW/20MWH Battery Storage. The Hyde Park BESS will be operated in parallel with Eversource's electrical grid. The BESS shall be comprised of a system of battery racks, inverters, control system, pad mounted transformers, and switchgear. The BESS will be located and interconnected to Eversource's existing 13.8 kV distribution infrastructure adjacent to the BESS facility.

The batteries would be dispatched with a peak load shave functionality. They will charge during the light load and discharge during the peak load in order to lower the substation peak. The substation peak exceeds the firm capacity of the station and is forecasted to keep rising due to step loads and EV charging load. The Battery can ride through multi-day peak load events. BESS can be connected to Baker St#110 side in the future through 25kV or 4kV feeders to support Baker Substation if needed, which will require underground cable construction.

Below is the preliminary distribution scope of work for the BESS interconnection to the four feeders.

Manhole Installation

-(3) concrete manholes with inside dimension of 6'x10'x8'

Conduit Installation

-(1,210') 6-5" PVC type EB encased in concrete from MH12362 to BESS via 3 new manholes

-(1,210') 6-5" PVC type EB encased in concrete from MH12362 to BESS via 3 new manholes Note: this second duct bank is required only if the substation is constructed

-(60') 6-5" PVC type EB encased in concrete from MH12362 to BESS via 3 new manholes

Primary Underground Cable Removal

-(13,065') of 15kV primary cable: 350/3, 3-350 FS, 3-500 FS, etc. (37 sections)

Primary Underground Cable Installation

-(16,395') of 3-700 FS 15kV primary cable (47 sections)

Below is the scope of work for connecting the BESS to feeder 496-H1XY.

- Cut and remove X-Y main from MH18478 to MH12361 and replace with 2,950 feet of 3-700 FS (9 sections) and make joints.

- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

- Note: 1 Single three-way joint in MH18478 and 3 single multiple joints in MH12361.

Below is the scope of work for connecting the BESS to feeder 441-1401H.

- Install new cable from MH12365 to MH12362 with 1,200 feet of 3-700 FS (3 sections) and make joints.

- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

Below is the scope of work for connecting the BESS to feeder 362-1207H.

- Cut and remove cable from MH15729 to MH12361 (5 sections).

- Install 1,810 feet of 3-500 FS (5 sections) from MH15729 to MH12361 and make joints.

- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

Below is the scope of work for connecting the BESS to feeder 362-1209H.

- Cut and remove cable from MH2434 to MH12358 (14 sections).

- Install 5,355 feet of 3-700 FS (14 sections) from MH2434 to MH12358 and make joints.

- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

To study the impact of the BESS on the Hyde Park distribution system, a transient dynamic analysis of the preferred alternative was performed in the PSCAD (Power Systems Computer Aided Design) tool. The four Hyde Park 13.8 kV distribution feeders along with the proposed BESS systems were modeled¹ in PSCAD by using circuit load and impedance data collected from steady-state Synergi models. The dynamic Risk of Islanding (ROI) analysis, and the Transient Overvoltage (TOV) analysis were done to verify compliance with IEEE 1547-2018 Standard.

The ROI analysis determined the BESS inverter's response to unintended island conditions. According to the IEEE standard, for an unintended islanding condition, the distributed energy resource (which in this case is the BESS) shall detect the island, cease to energize the island, and trip within 2 seconds of the formation of the island.

¹ The BESS systems on the distribution circuits were modelled with a generic inverter model, not specific to any one manufacturer. After a vendor for the BESS is selected, the dynamic study will have to be re-run with the manufacturer specific inverter model.

When interconnecting distributed energy resource there is a concern that during step changes in load (such as tripping of an upstream device), the proposed inverters may cause transient over voltages in excess of 1.2 per unit, which can potentially cause damage to the customer's equipment and/or utility equipment. The TOV analysis was conducted to verify if the expected transient overvoltage conditions caused by the BESS inverters during step changes in load were within standard limits.

The ROI and the TOV analyses were performed on the normal (N-0) configuration and several contingency (N-1) configuration scenarios. The N-1 operational scenarios included one of the four feeder outages, a transformer outage, and a bus outage. The BESS system passed the ROI test for all scenarios tested. The BESS system also does not contribute to any TOV conditions for any of the tested scenarios.

Risk of Islanding (ROI) Screening

Based on the suggested guidelines for anti-islanding screening from Sandia National Laboratories and Eversource's distributed energy resources (DER) screening for ROI assessment (shown in Figure 1), additional requirements of ROI are necessary. Further study was performed to determine if the BESS inverters are reacting appropriately and not cause violations.

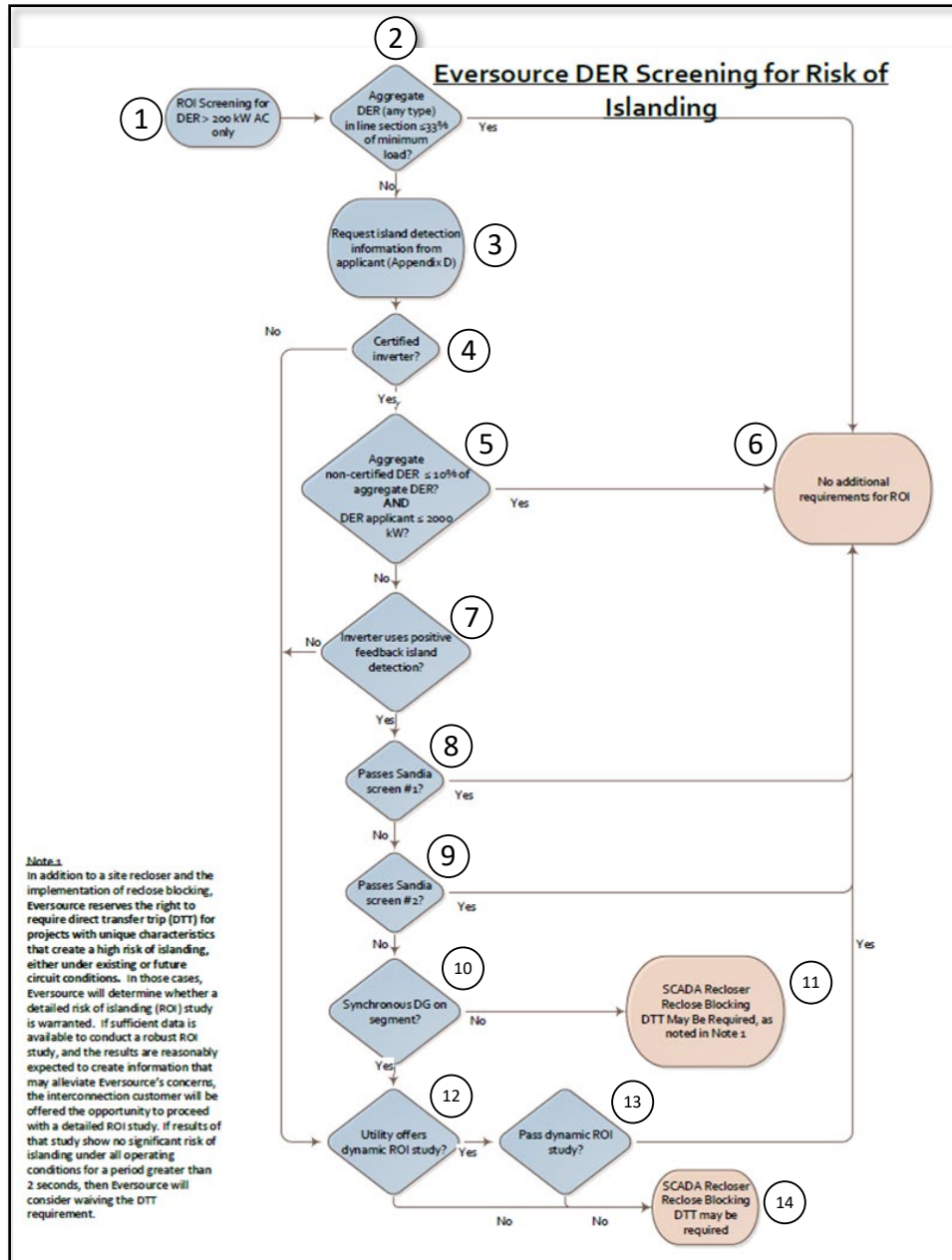


FIGURE 1. RISK OF ISLANDING (ROI) SCREENING PROCESS

During the dynamic study, in PSCAD, a detailed risk of islanding analysis was performed. The BESS system passed the ROI test for all four feeder interconnections.

Feeder	DER Interconnection	ROI Test
441-1401H	2.5 MW BESS	PASS
496-H1XY	2.5 MW BESS	PASS
362-1207H	2.5 MW BESS	PASS
362-1209H	2.5 MW BESS	PASS

TABLE 1: HYDE PARK BESS DYNAMIC STUDY – ROI RESULTS

PSCAD analysis shows that the BESS inverter trips within 2 seconds of unintentional island formation.

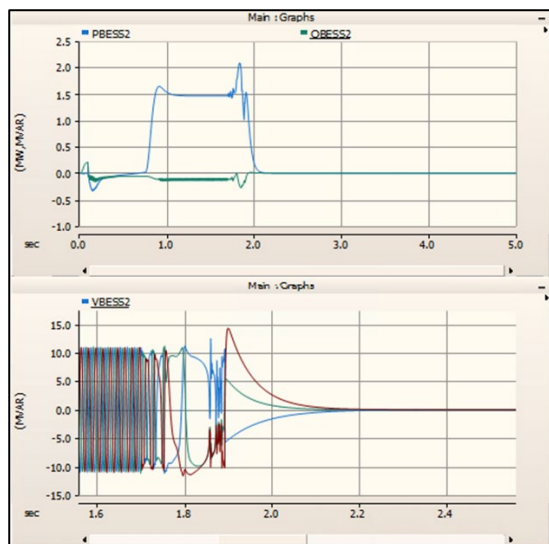


FIGURE 2: ROI TEST FOR FEEDER 441-1401H

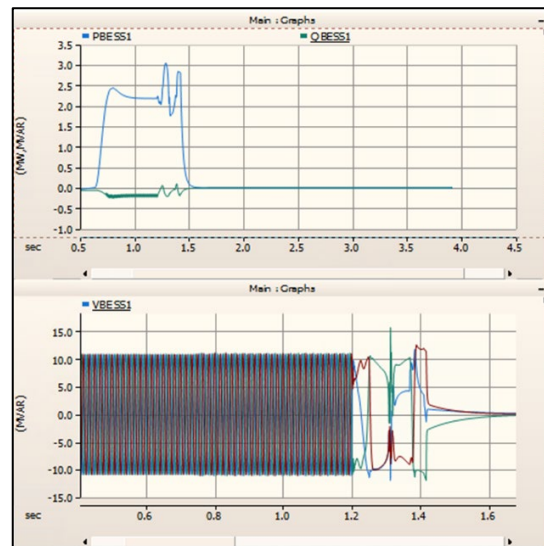


FIGURE 3: ROI TEST FOR FEEDER 496-H1XY

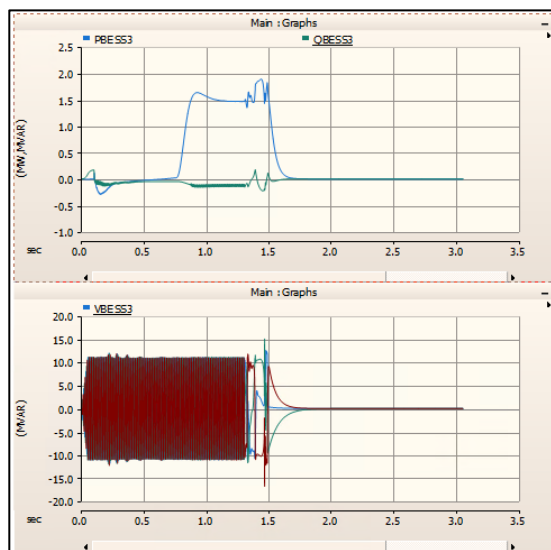


FIGURE 4: ROI TEST FOR FEEDER 362-1207H

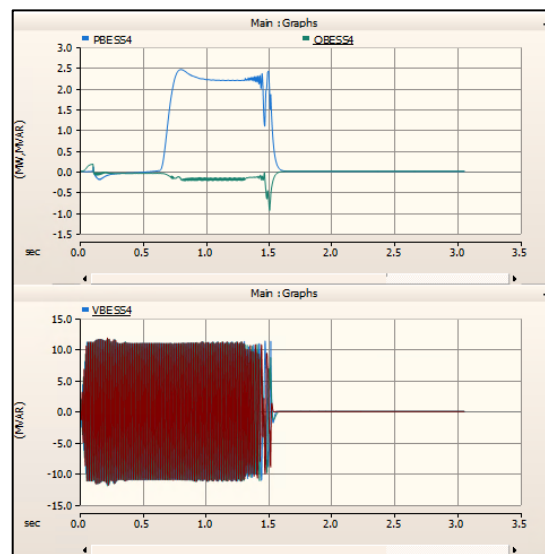


FIGURE 5: ROI TEST FOR FEEDER 362-1209H

Transient Overvoltage (TOV) Screening

If aggregate DER is greater than or equal to 115% of minimum gross load of the circuit, then there is a risk of Transient Overvoltage (TOV) at the feeder level and/or substation level. The same PSCAD dynamic study required for Risk of Islanding is further necessary to evaluate the TOV concerns. For dynamic transient overvoltage study, circuit load and impedance data are collected from steady-state Synergi models and used to build dynamic feeder models. The BESS inverters were modelled using generic PSCAD models. All other generators on the distribution circuits are modelled with a generic inverter model, not specific to any one manufacturer. The TOV analysis includes both Load Rejection Overvoltage (LROV) and Ground Fault Overvoltage (GFOV) studies.

Load Rejection Overvoltage (LROV) occurs when an upstream circuit breaker or recloser opens, which can cause an overvoltage on the isolated circuit due to stored energy in line and transformer inductances. LROV is modelled in PSCAD by operating all generation at 100% output in a minimum load case and opening the feeder circuit breaker. The voltage at the Point of Interconnection (POI) is measured.

Ground Fault Overvoltage (GFOV) is simulated by using the same minimum load, maximum generation scenario as the TOV scenario. A single-line-to-ground fault is initiated at the high side of the BESS interconnection transformer which persists until the upstream feeder breaker interrupts the fault after 10 cycles (0.16667 seconds). This is a conservative simulation and models a failure of protection selectivity.

Both LROV and GFOV durations must not exceed the following limits specified in IEEE 1547.

Transient Overvoltage Magnitude (per unit)	Maximum Allowable Duration (ms)
1.2	166
1.3	16
1.4	3
1.7	1.6
2.0	0

TABLE 2: IEEE 1547 STANDARD

Both, the LROV and GFOV analyses were performed on the normal (N-0) configuration and several contingency (N-1) configuration scenarios which included any one element out of service (OOS). The N-1 operational scenarios included:

- i. Transformer #1 OOS
- ii. Bus #1 OOS
- iii. Bus #3 OOS

The normal configuration of the proposed BESS at Hyde Park station is shown on the one-line diagram in figure 6. The N-1 contingency scenarios and corresponding breaker operations are shown in table 3.

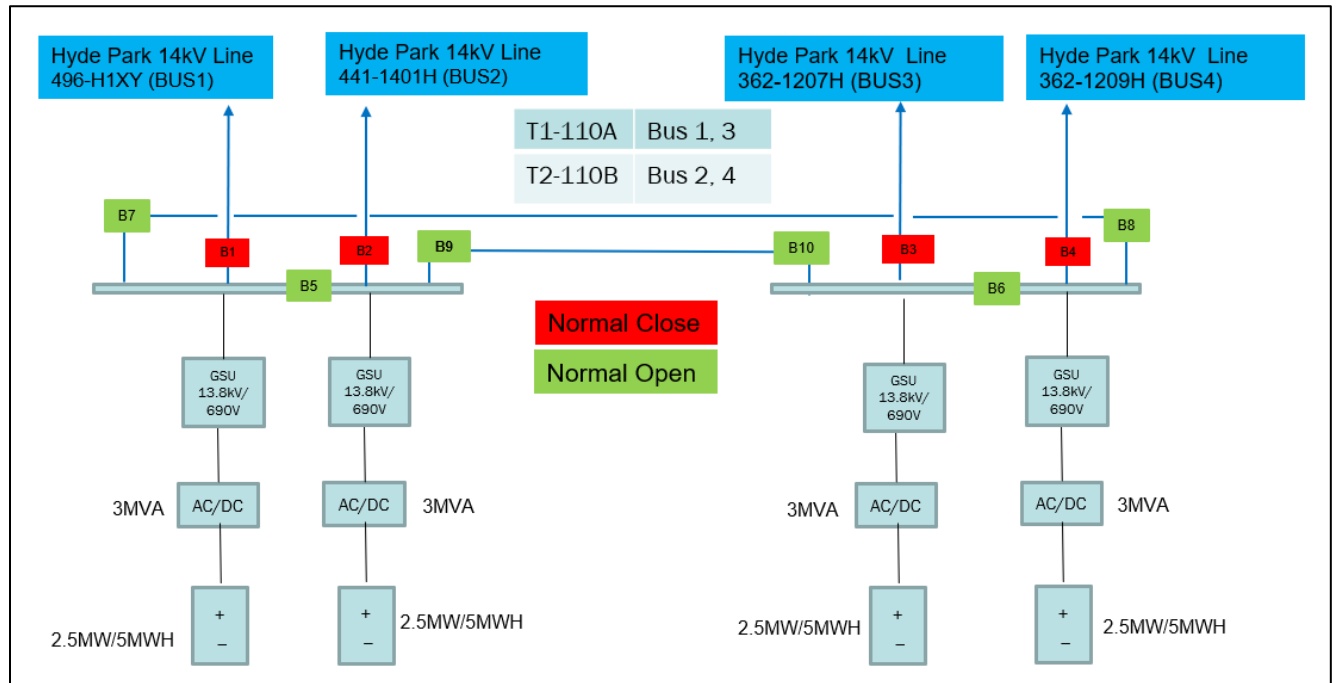


FIGURE 6: HYDE PARK PROPOSED BESS ONE-LINE DIAGRAM

Scenarios	Close	Open
Normal	B1, B2, B3, B4	B5, B6, B7, B8, B9, B10
T1- 110A OOS	B2, B4, B5, B6	B1, B3, B7, B8, B9, B10
Bus #1 OOS	B2, B3, B4, B5	B1, B6, B7, B8, B9, B10
Bus #3 OOS	B1, B2, B4, B6	B3, B5, B7, B8, B9, B10

TABLE 3: LIST OF TEST SCENARIOS

Results at Project (BESS) POI under Normal Configuration (N-0):

Load Rejection Overvoltage (LROV) Results – Feeder 441-1401H BESS Interconnection

BESS does not contribute to load rejection overvoltage on feeder 441-1401H.

Pre-project				
Feeder 441-1401H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0
1.3 pu	16	0	0	0
1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0

Post-project				
Feeder 441-1401H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0.56	0
1.3 pu	16	0	0	0
1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0

Ground Fault Overvoltage (GFOV) Results – Feeder 441-1401H BESS Interconnection

BESS does not contribute to ground fault overvoltage on feeder 441-1401H.

Pre-project				
Feeder 441-1401H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0
1.3 pu	16	0	0	0
1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0

Post-project				
Feeder 441-1401H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0
1.3 pu	16	0	0	0
1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0

Load Rejection Overvoltage (LROV) Results – Feeder 496-H1XY BESS Interconnection

BESS does not contribute to load rejection overvoltage on feeder 496-H1XY.

Pre-project					Post-project				
Feeder 496-H1XY					Feeder 496-H1XY				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	2.10	1.80	0
1.3 pu	16	0	0	0	1.3 pu	16	1.66	1.36	0
1.4 pu	3	0	0	0	1.4 pu	3	1.26	0.98	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0.22	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Ground Fault Overvoltage (GFOV) Results – Feeder 496-H1XY BESS Interconnection

BESS does not contribute to ground fault overvoltage on feeder 496-H1XY.

Pre-project					Post-project				
Feeder 496-H1XY					Feeder 496-H1XY				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	0	24.54
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0.52
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Load Rejection Overvoltage (LROV) Results –Feeder 362-1207H Interconnection

BESS does not contribute to load rejection overvoltage on feeder 362-1207H.

Pre-project					Post-project				
Feeder 362-1207H					Feeder 362-1207H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0.20	0	0.02
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0.02
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Ground Fault Overvoltage (GFOV) Results –Feeder 362-1207H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1207H.

Pre-project					Post-project				
Feeder 362-1207H					Feeder 362-1207H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	36.76	0	0
1.3 pu	16	0	0	0	1.3 pu	16	1.42	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Load Rejection Overvoltage (LROV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to load rejection overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	0	0
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Ground Fault Overvoltage (GFOV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	25.84	0
1.3 pu	16	0	0	0	1.3 pu	16	0	7.88	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Results at Project (BESS) POI under several Contingency Configurations (N-1):

(i) Transformer #1 Out of Service:

Load Rejection Overvoltage (LROV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	0	0.02
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Ground Fault Overvoltage (GFOV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0.04	4.42	0
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

(ii) Bus #1 Out of Service:

Load Rejection Overvoltage (LROV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	0.08	0
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

(iii) Bus #3 Out of Service:

Ground Fault Overvoltage (GFOV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0.02	4.20	0
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Steady State Analysis Summary Results

Power Flow Study

The below graphs show that the four (4) 2.5MW/5 MWh battery units can reduce the peak load to approximately 95% of the Load Carrying Capacity of the Hyde Park Substation in year 2032 or approximately 148 MVA. The BESS would be interconnected to four (4) 14kV distribution feeders, 496-H1XY, 441-1401, 362-1209H and 362-1207H. One 2.5MW/5 MWh battery will be connected to each of the four feeders.

The project is designed to meet the 10 year forecast, where the highest load being is in 2032. Year 2032 is chosen as the year with the highest load forecast of 155 MVA for the Hyde Park Substation without any batteries installed. The batteries act as a load and charge from 04:00 **AM** to 09:00 AM and discharge from 17:00 PM to 21:00 PM to decrease the Substation peak load as shown below.

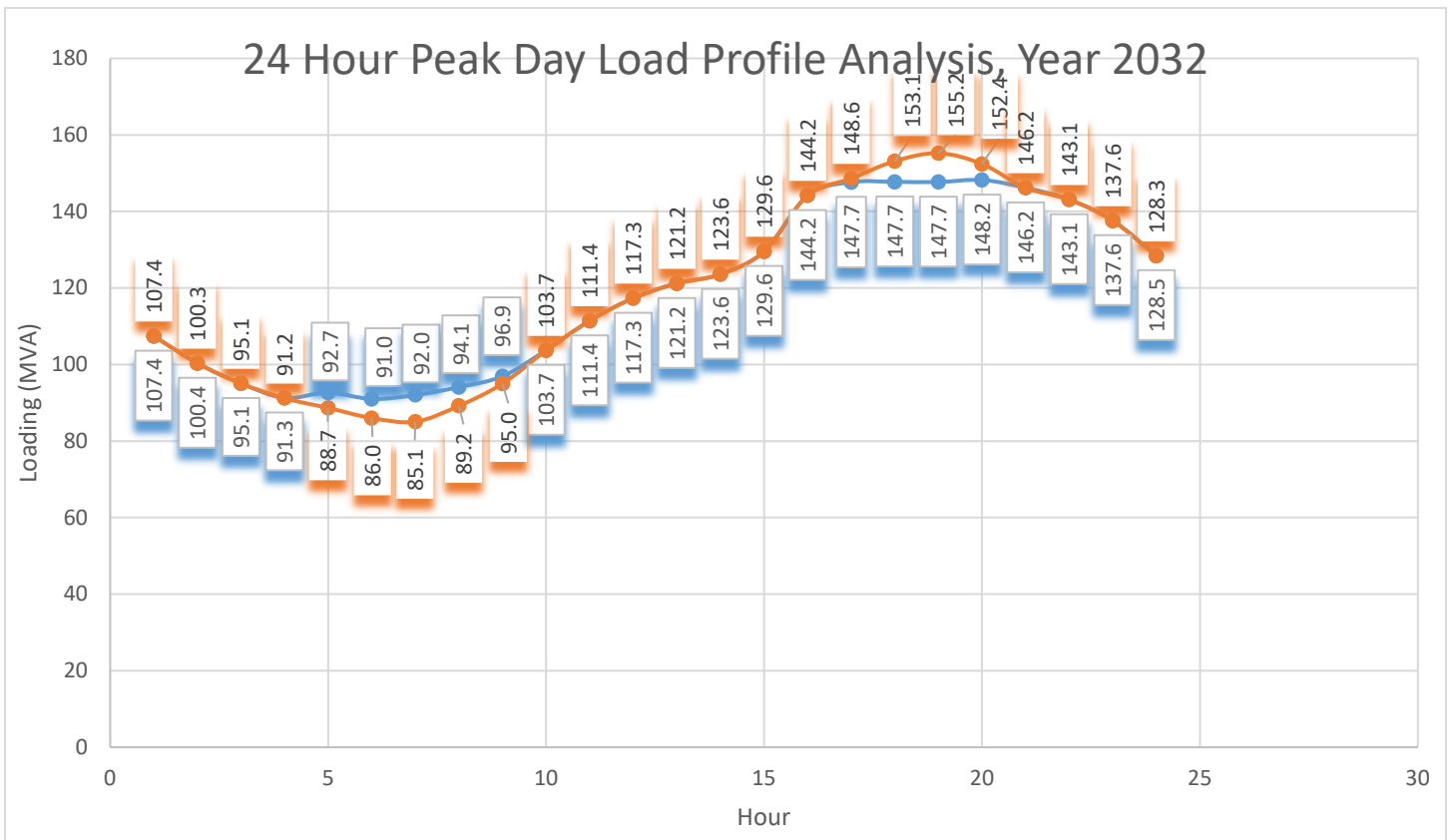


Figure 7

The orange line represents the Hyde Park Substation load without any batteries installed and the blue line represents the Substation load with the four batteries installed.

The below graph shows the resultant load profile for the substation during the peak load day under contingency assuming 13.17 MVA of distribution load transfers and the battery is operational. This decrease load to 141 MVA is due to a load transfer of 13.17 MVA which occurs during N-1 condition through distribution feeder ties and the decrease to 133.7 at 19:00 is due to the BESS.

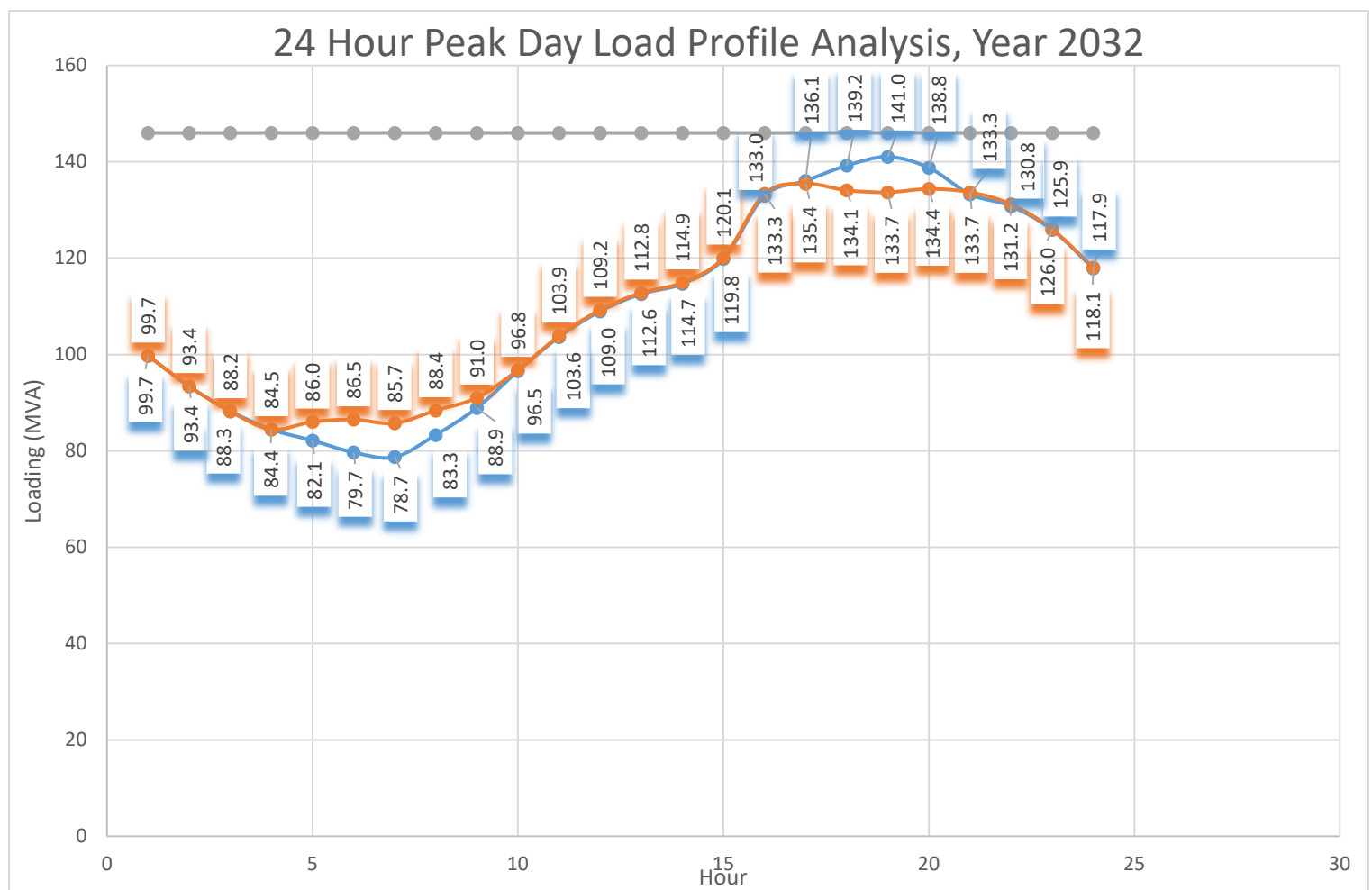


Figure 8

Thermal Rating Analysis

Below graph shows the 2032 24 hour peak day loading on each of the four feeders that the BESS is interconnected to and assuming the BESS is operational. It shows the 24 hour peak load

day profile for the four selected feeders in MVA based on the 2032 modeled loads. The batteries act as a load and charge from 04:00 to 09:00 and discharge from 17:00 to 21:00 to decrease the Substation peak load.

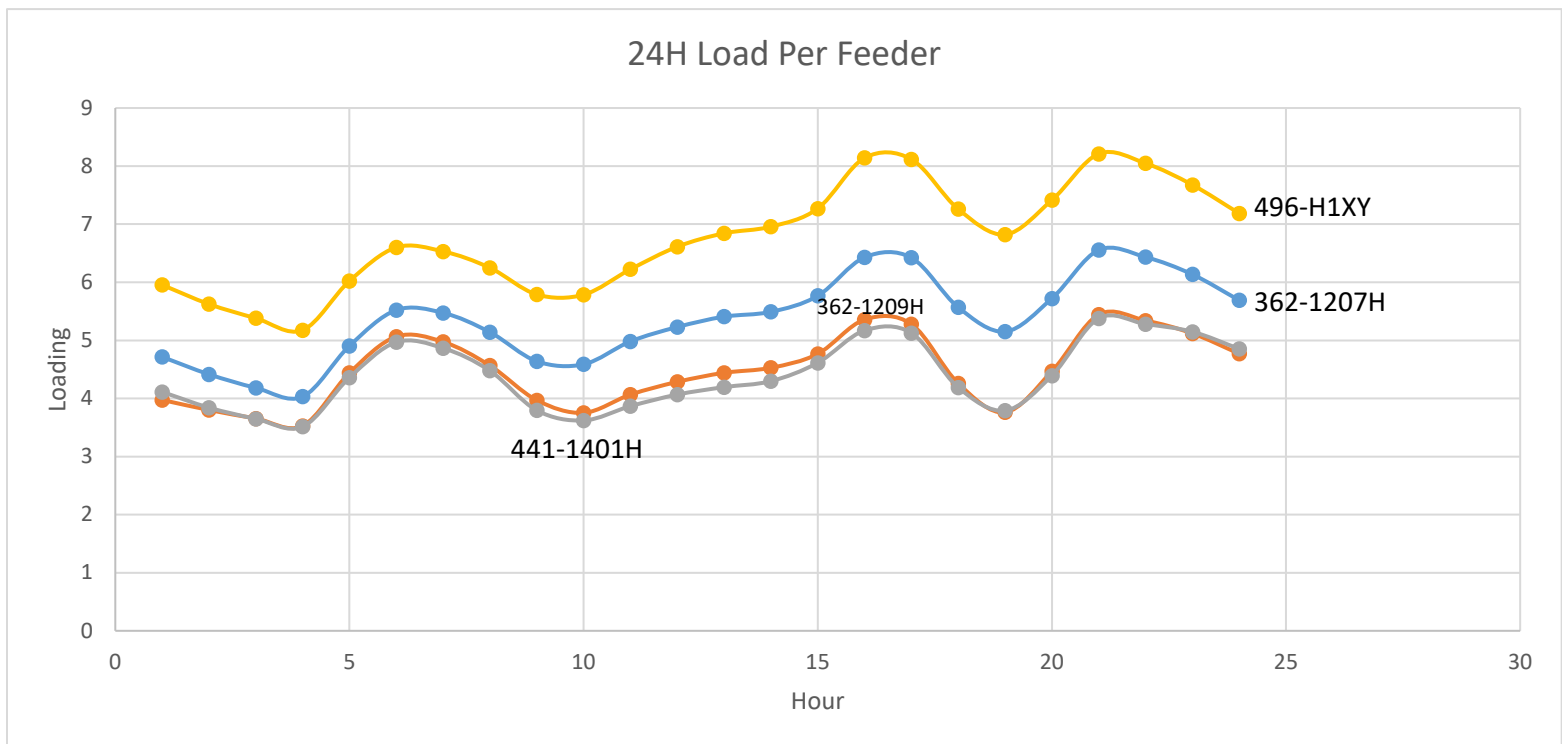


Figure 9

As shown by the below table, the circuit loading is below 80 percent of the circuit ratings when the battery is charging from 4 AM to 9 AM and acting as a load.

	Feeder_362_1207H	Feeder_362_1209H	Feeder_441_1401H	Feeder_496_H1XY
5:00 AM	4.9	4.4	4.4	6.0
6:00 AM	5.5	5.1	5.0	6.6
7:00 AM	5.5	5.0	4.9	6.5
8:00 AM	5.1	4.6	4.5	6.3
9:00 AM	4.6	4.0	3.8	5.8
80% of Circuit Rating (MVA)	6.72	6.72	8.72	7.36
Circuit Rating (MVA)	8.4	8.4	10.9	9.2

The exact loading for the year 2032 cannot be predicted at this stage. However, based on the above loading data, there is a flexibility to charge the BESS on three of the above feeders instead of four if need be. Based on the above loading, feeder 441-1401 has the lowest amount of load and would be able to accommodate another 2.5 MW of BESS while feeder 496-H1XY may not. System Planning will be monitoring and modifying the battery schedule based on the EV charging profile and its load on the Hyde Park distribution circuits.

Below is a graph showing circuit loading for each of the four feeders under a Hyde Park Substation transformer contingency with the BESS connected to each of the four feeders. It shows the 24 hour peak load day profile for the four selected feeders in MVA based on the 2032 modeled loads. The batteries act as a load and charge from 04:00 to 09:00 and discharge from 17:00 to 21:00 to decrease the Substation peak load.

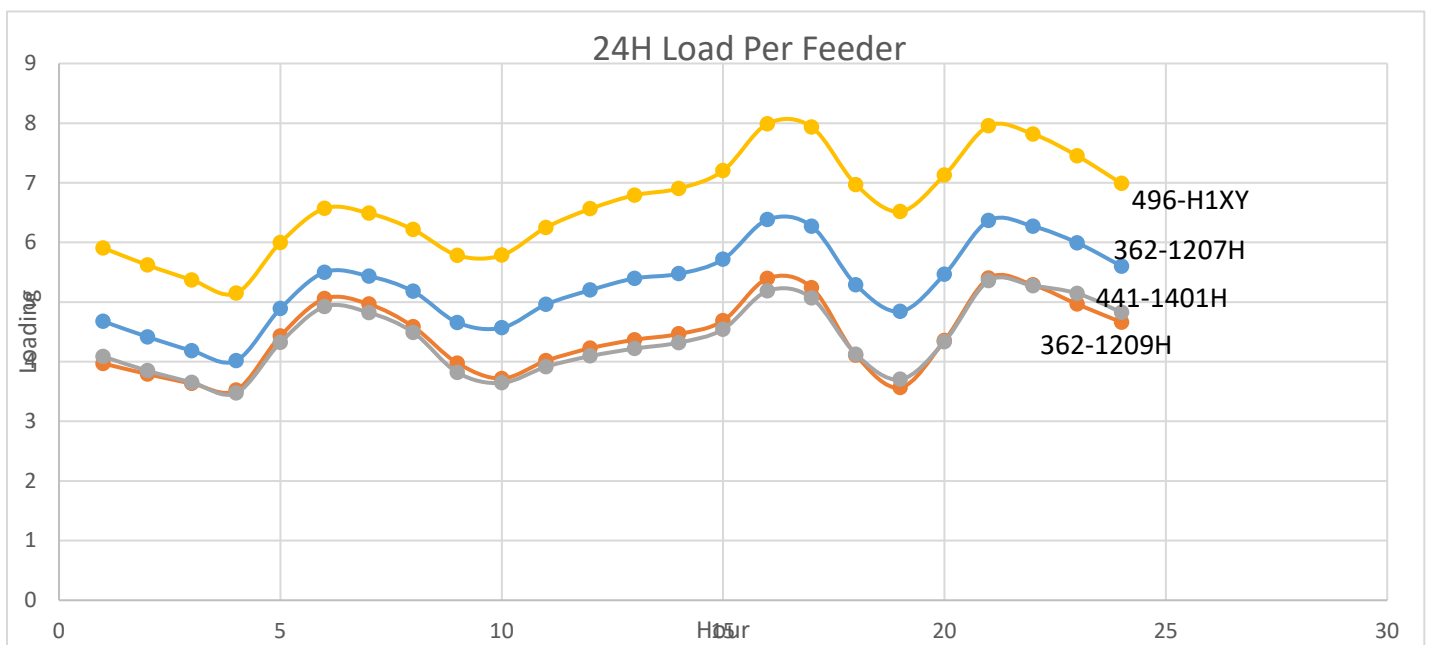


Figure 10

As shown by the below table, the circuit loading is below 80 percent of the circuit ratings when the battery is charging from 4 AM to 9 AM and acting as a load.

	Feeder_362_1207H	Feeder_362_1209H	Feeder_441_1401H	Feeder_496_H1XY
5:00 AM	4.89	4.43	4.32	5.996

6:00 AM	5.50	5.06	4.92	6.57
7:00 AM	5.43	4.96	4.82	6.49
8:00 AM	5.18	4.59	4.49	6.21
9:00 AM	4.66	3.97	3.82	5.783
80% of Circuit Rating (MVA)	6.72	6.72	8.72	7.36
Circuit Rating (MVA)	8.4	8.4	10.9	9.2

The BESS also helps to bring down the peak load on these 14 kV feeders. The Hyde Park circuit 496-H1XY historically runs 80-90% of its summer normal ampacity in the summer. The circuit 362-1207 supplied historically runs 60-80% of its summer normal in the summer months. The circuit 362-1209 historically runs 60-90% of its summer normal in the summer months.

Reverse Power Flow Evaluation

Due to the small size of the battery, there is no reverse power flow into the Hyde Park substation transformers. To evaluate the reverse power flow, the worst case scenario was analyzed where two batteries with a total size of 5MW/10MWH were connected to one feeder and the BESS was operating at a 100% output. This is assuming there is a contingency on one of the four 14 kV feeders and there would need to be two 2.5MW/5MWH BESS on one feeder. It is seen that the load on the transformer supplying that feeder had just decreased by 5 MW due to the battery. Therefore, there is no reverse power flow because the battery would just decrease the load and is not large enough to cause reverse power flow into the electric grid.

For example, the below tables show the loading on the transformer supplying feeder 496-H1XY before and after 5MW/10MWH BESS was connected to it and operating at 100%output.

Transformer: STA_496_110A_2

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Transformer	View analysis results					
LTC / LDC	Result	A/AB	B/BC	C/CA	Avg	Mx/T
Settings	▼ General					
Tertiary	Ft Into	495578	465396	485737	482237	495578
Net Protector	Ft Away	---	---	---	---	0
Rates	Nom kV	ABC	ABCN	---	---	13.80
Info	▼ Taps and LTC					
Results	Vrr	---	121.7	---	121.7	---
	Amps Out	1599	1611	1615	1608	1615
	Tap Pos	11	11	11	11	11
	▼ Volts and Amps					
	Cumul Volt Drop	0.0	0.0	0.0	0.0	0.0
	Volts Out	121.8	121.7	121.7	121.8	121.8
	Amps Into	206	206	207	206	207
	Pct pf	92.0	92.0	92.0	92.0	---
	Imbalance	V,I,In	0.0%	0.3%	0 A	---
	▼ Power Flow					
	MVA Into	13.70	13.70	13.70	13.70	41.10
	MW Into	12.60	12.60	12.60	12.60	37.80
	Mvar Into	5.38	5.38	5.38	5.38	16.14
	MW Demand	12.46	12.55	12.58	12.53	37.59
	Mvar Demand	3.46	3.47	3.47	3.47	10.40
	Nom kvar Into	D,S,LC	0	0	0	---
	Nom kW Into	D,S,LC	0	0	0	---

Power Flow Without BESS

Transformer: STA_496_110A_2



Transformer	View analysis results
LTC / LDC	
Settings	
Tertiary	
Net Protector	
Rates	
Info	
Results	

Result	A/AB	B/BC	C/CA	Avg	Mx/T
General					
Ft Into	495578	465396	485737	482237	495578
Ft Away	---	---	---	---	0
Nom kV	ABC	ABCN	---	---	13.80
Taps and LTC					
Vrr	---	121.8	---	121.8	---
Amps Out	1412	1424	1427	1421	1427
Tap Pos	11	11	11	11	11
Volts and Amps					
Cumul Volt Drop	0.0	0.0	0.0	0.0	0.0
Volts Out	121.8	121.8	121.8	121.8	121.8
Amps Into	182	182	183	182	183
Pct pf	90.4	90.4	90.4	90.4	---
Imbalance	V,I,In	0.0%	0.3%	0 A	---
Power Flow					
MVA Into	12.11	12.11	12.11	12.11	36.32
MW Into	10.94	10.94	10.94	10.94	32.83
Mvar Into	5.18	5.18	5.18	5.18	15.55
MW Demand	10.81	10.91	10.94	10.89	32.66
Mvar Demand	3.69	3.69	3.69	3.69	11.07
Nom kvar Into	D,S,LC	0	0	0	---
Nom kW Into	D,S,LC	0	0	0	---

Power Flow With BESS

As shown, the load on the transformer supplying that feeder had just decreased by 5 MW due to the battery and there is no reverse power flow.

Flicker

The flicker values comply with the Eversource's allowable limit of 2.0% at the distribution circuits. The voltage difference when the BESS is off and when the BESS is operating at 100% output was analyzed and was below 2% during normal and contingency scenarios. The worst case scenario where there are two BESS with a total size of 5 MW/10MWH operating at 100% output on one feeder was also analyzed. The voltage difference still remained below 2%. This voltage difference was analyzed at the peak load of 155 MVA.

Transformer LTC Impact

The tap position range from -16 to 16 and the voltage is maintained at 122 Volts at the 14 kV buses. When the BESS is connected, the voltage increases slightly due to the decrease in load. Therefore, the BESS may cause the LTC to move to a slightly lower tap position to maintain the voltage at 122 volts. Overall, due to the small size of the battery, it would not have much impact on the transformer LTC tap positions.

Short Circuit

Due to the small size of the battery, it would not have much impact on the short circuit current.